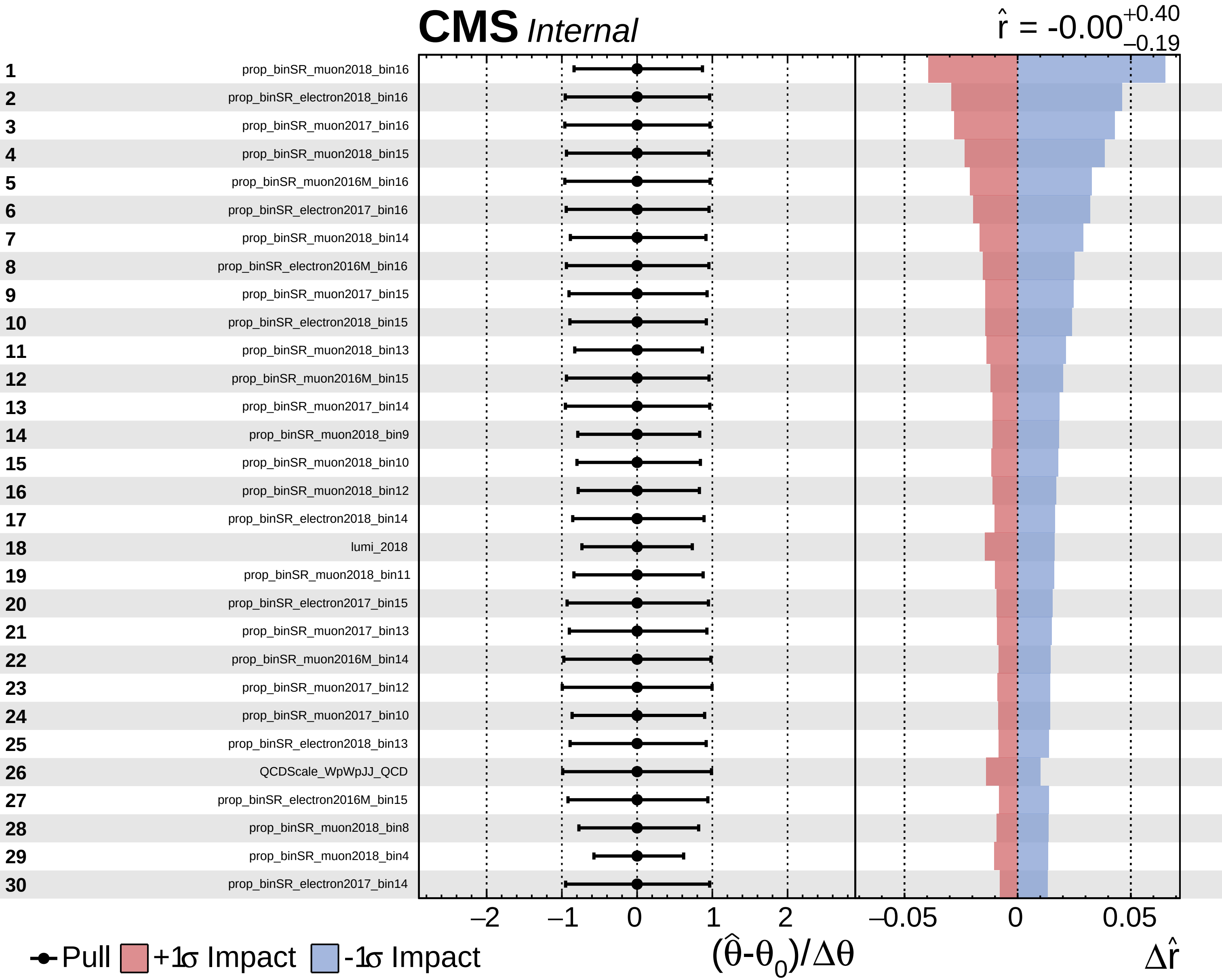
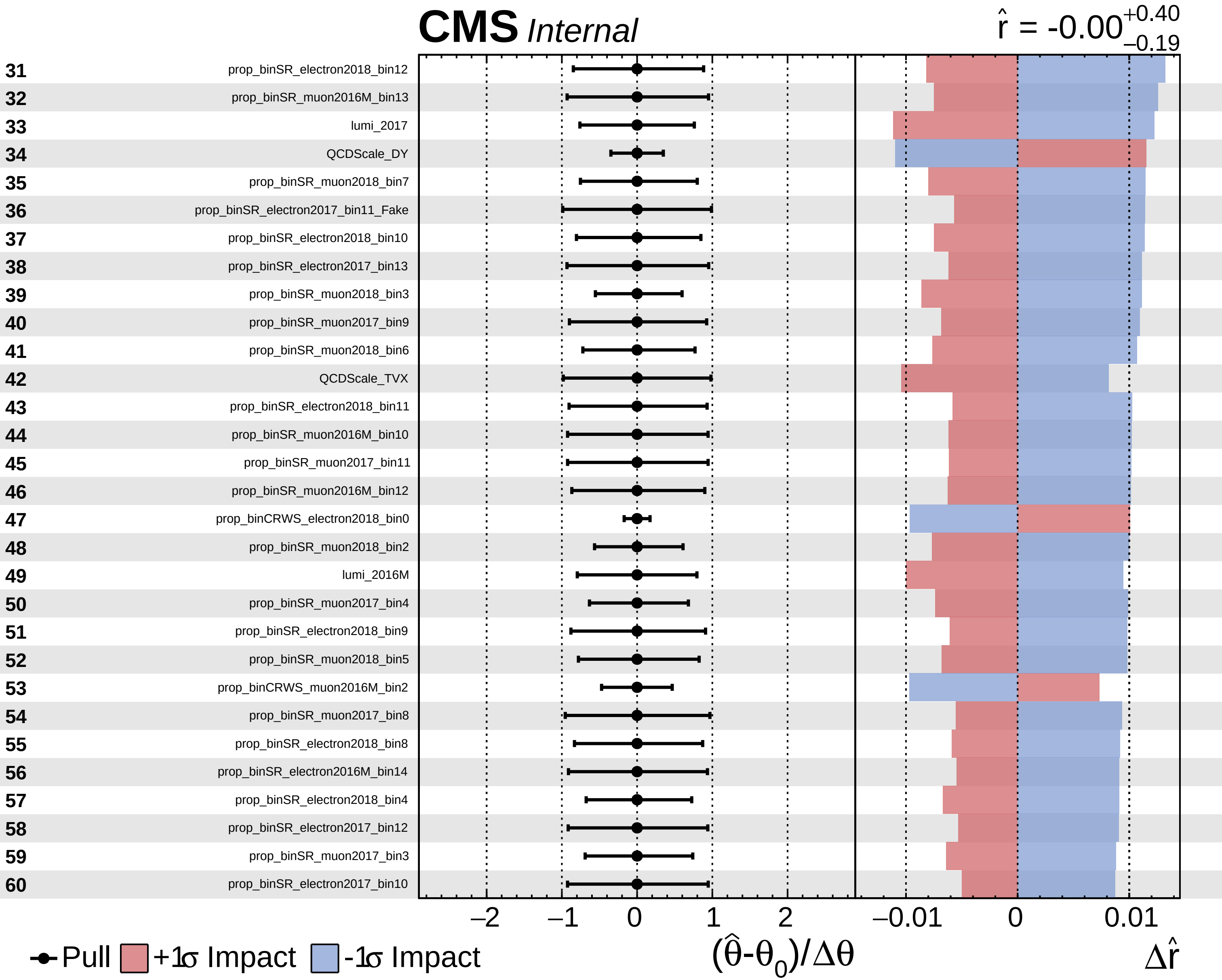
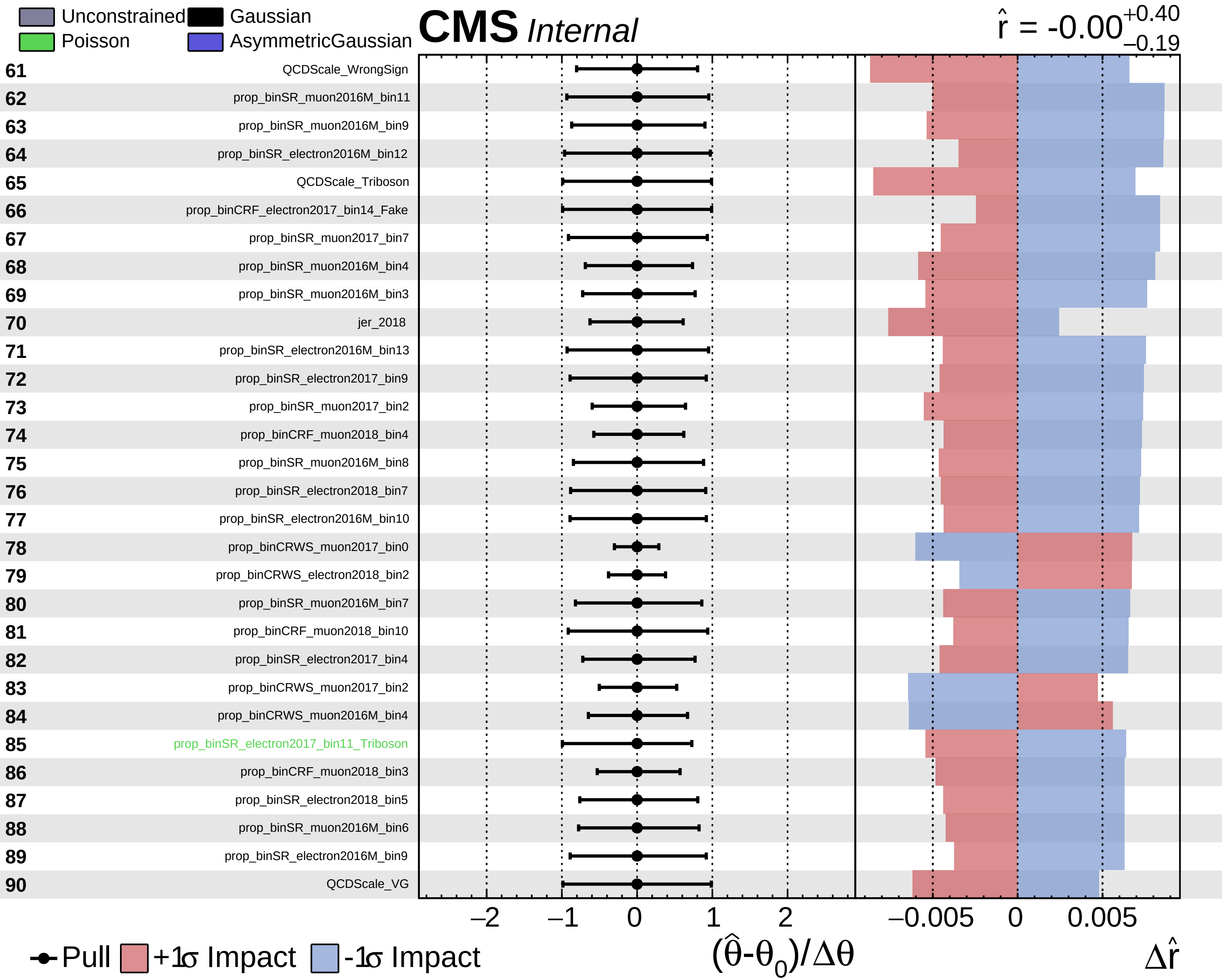


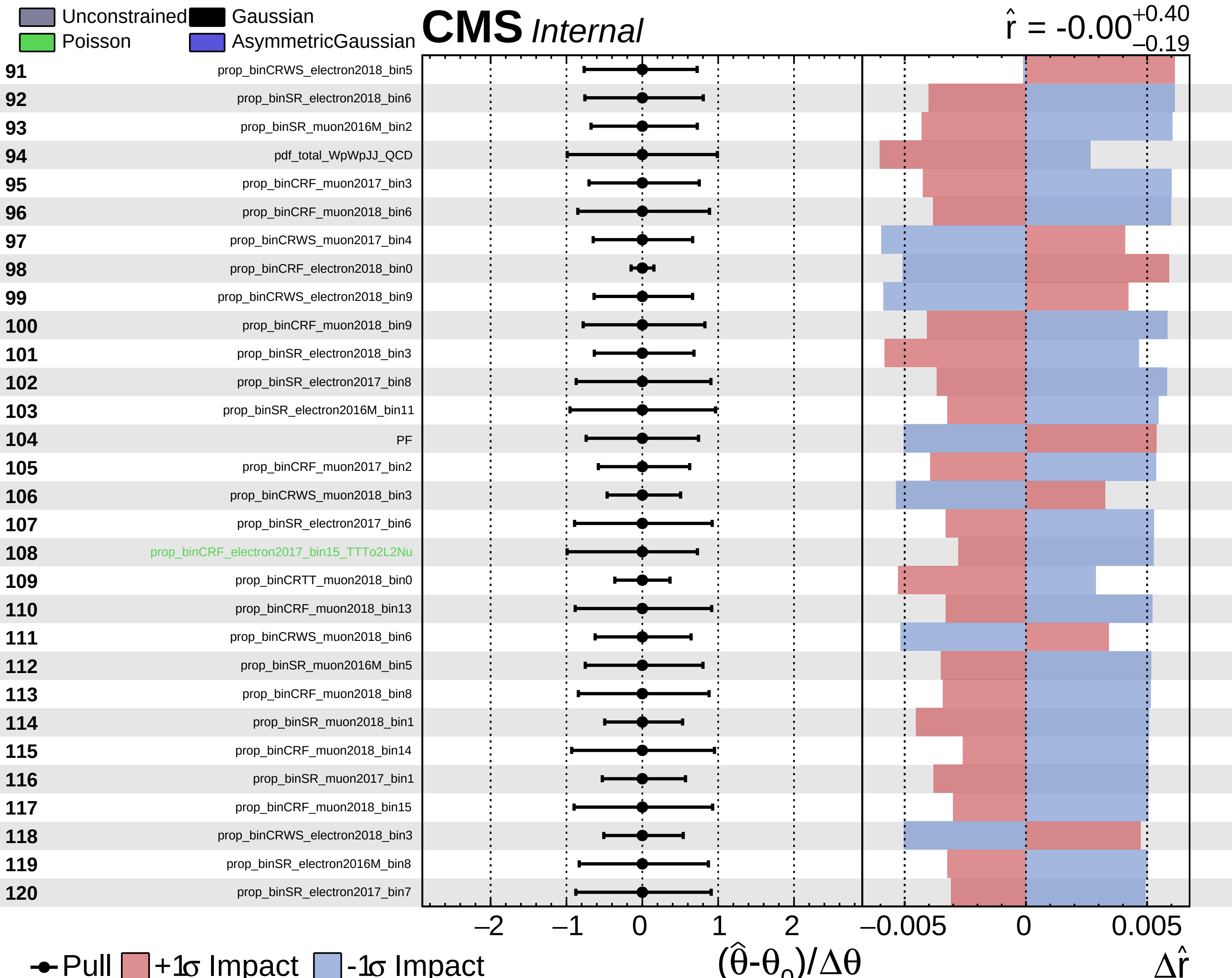
CMS Internal
$\hat{r} = -0.00^{+0.40}_{-0.19}$
1 prop_binSR_muon2018_bin16
2 prop_binSR_electron2018_bin16
3 prop_binSR_muon2017_bin16
4 prop_binSR_muon2018_bin15
5 prop_binSR_muon2016M_bin16
6 prop_binSR_electron2017_bin16
7 prop_binSR_muon2018_bin14
8 prop_binSR_electron2016M_bin16
9 prop_binSR_muon2017_bin15
10 prop_binSR_electron2018_bin15
11 prop_binSR_muon2018_bin13
12 prop_binSR_muon2016M_bin15
13 prop_binSR_muon2017_bin14
14 prop_binSR_muon2018_bin9
15 prop_binSR_muon2018_bin10
16 prop_binSR_muon2018_bin12
17 prop_binSR_electron2018_bin14
18 lumi_2018
19 prop_binSR_muon2018_bin11
20 prop_binSR_electron2017_bin15
21 prop_binSR_muon2017_bin13
22 prop_binSR_muon2016M_bin14
23 prop_binSR_muon2017_bin12
24 prop_binSR_muon2017_bin10
25 prop_binSR_electron2018_bin13
26 QCDScale_WpWpJJ_QCD
27 prop_binSR_electron2016M_bin15
28 prop_binSR_muon2018_bin8
29 prop_binSR_muon2018_bin4
30 prop_binSR_electron2017_bin14
−2 −1 0 1 2
−0.05 0 0.05
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



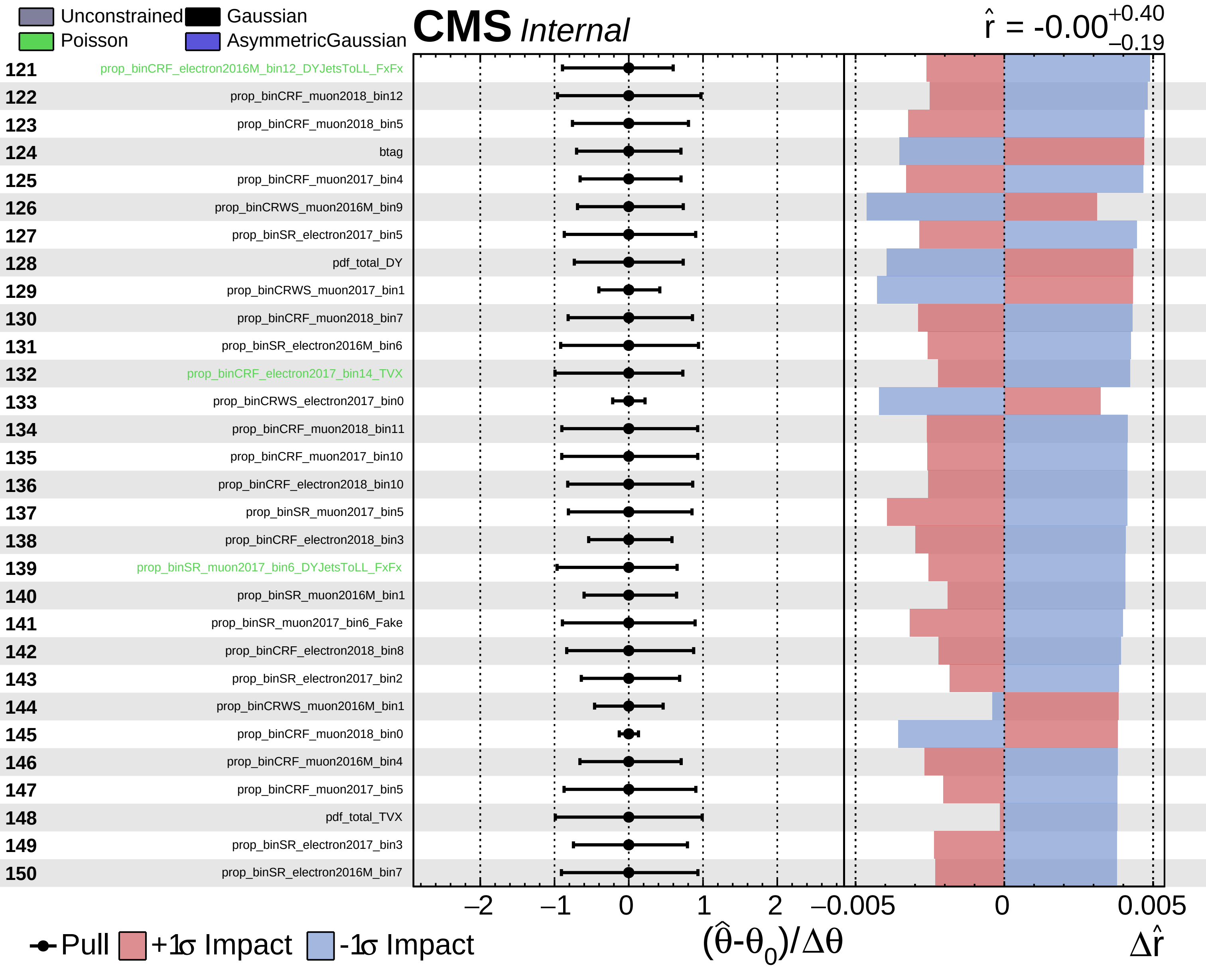
CMS Internal
r̂ = -0.00 +0.40 −0.19
31 prop_binSR_electron2018_bin12
32 prop_binSR_muon2016M_bin13
33 lumi_2017
34 QCDScale_DY
35 prop_binSR_muon2018_bin7
36 prop_binSR_electron2017_bin11_Fake
37 prop_binSR_electron2018_bin10
38 prop_binSR_electron2017_bin13
39 prop_binSR_muon2018_bin3
40 prop_binSR_muon2017_bin9
41 prop_binSR_muon2018_bin6
42 QCDScale_TVX
43 prop_binSR_electron2018_bin11
44 prop_binSR_muon2016M_bin10
45 prop_binSR_muon2017_bin11
46 prop_binSR_muon2016M_bin12
47 prop_binCRWS_electron2018_bin0
48 prop_binSR_muon2018_bin2
49 lumi_2016M
50 prop_binSR_muon2017_bin4
51 prop_binSR_electron2018_bin9
52 prop_binSR_muon2018_bin5
53 prop_binCRWS_muon2016M_bin2
54 prop_binSR_muon2017_bin8
55 prop_binSR_electron2018_bin8
56 prop_binSR_electron2016M_bin14
57 prop_binSR_electron2018_bin4
58 prop_binSR_electron2017_bin12
59 prop_binSR_muon2017_bin3
60 prop_binSR_electron2017_bin10
−2 −1 0 1 2
−0.01 0 0.01
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δr̂



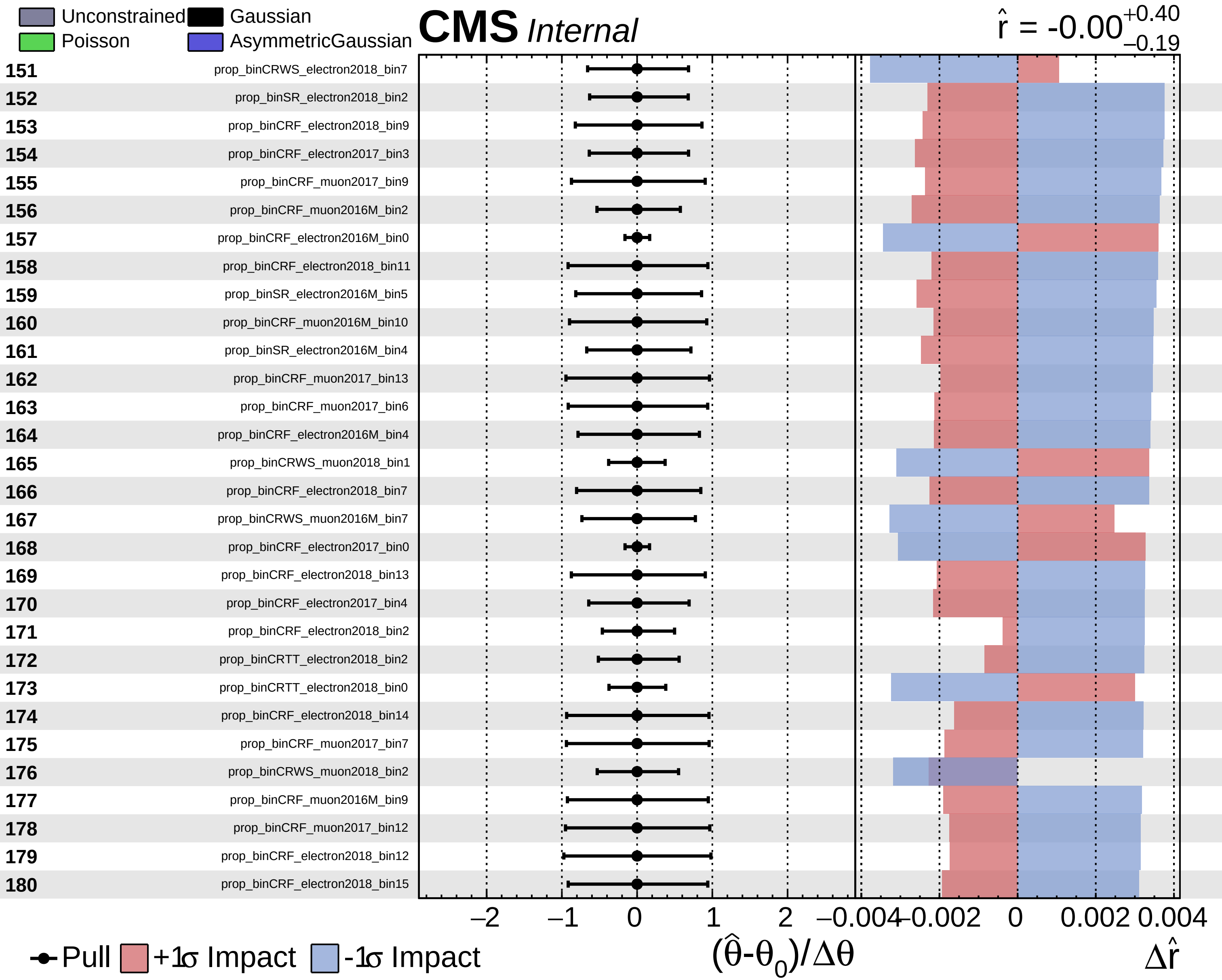
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.40}_{-0.19}$
61 QCDScale_WrongSign
62 prop_binSR_muon2016M_bin11
63 prop_binSR_muon2016M_bin9
64 prop_binSR_electron2016M_bin12
65 QCDScale_Triboson
66 prop_binCRF_electron2017_bin14_Fake
67 prop_binSR_muon2017_bin7
68 prop_binSR_muon2016M_bin4
69 prop_binSR_muon2016M_bin3
70 jer_2018
71 prop_binSR_electron2016M_bin13
72 prop_binSR_electron2017_bin9
73 prop_binSR_muon2017_bin2
74 prop_binCRF_muon2018_bin4
75 prop_binSR_muon2016M_bin8
76 prop_binSR_electron2018_bin7
77 prop_binSR_electron2016M_bin10
78 prop_binCRWS_muon2017_bin0
79 prop_binCRWS_electron2018_bin2
80 prop_binSR_muon2016M_bin7
81 prop_binCRF_muon2018_bin10
82 prop_binSR_electron2017_bin4
83 prop_binCRWS_muon2017_bin2
84 prop_binCRWS_muon2016M_bin4
85 prop_binSR_electron2017_bin11_Triboson
86 prop_binCRF_muon2018_bin3
87 prop_binSR_electron2018_bin5
88 prop_binSR_muon2016M_bin6
89 prop_binSR_electron2016M_bin9
90 QCDScale_VG
−2 −1 0 1 2
−0.005 0 0.005
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



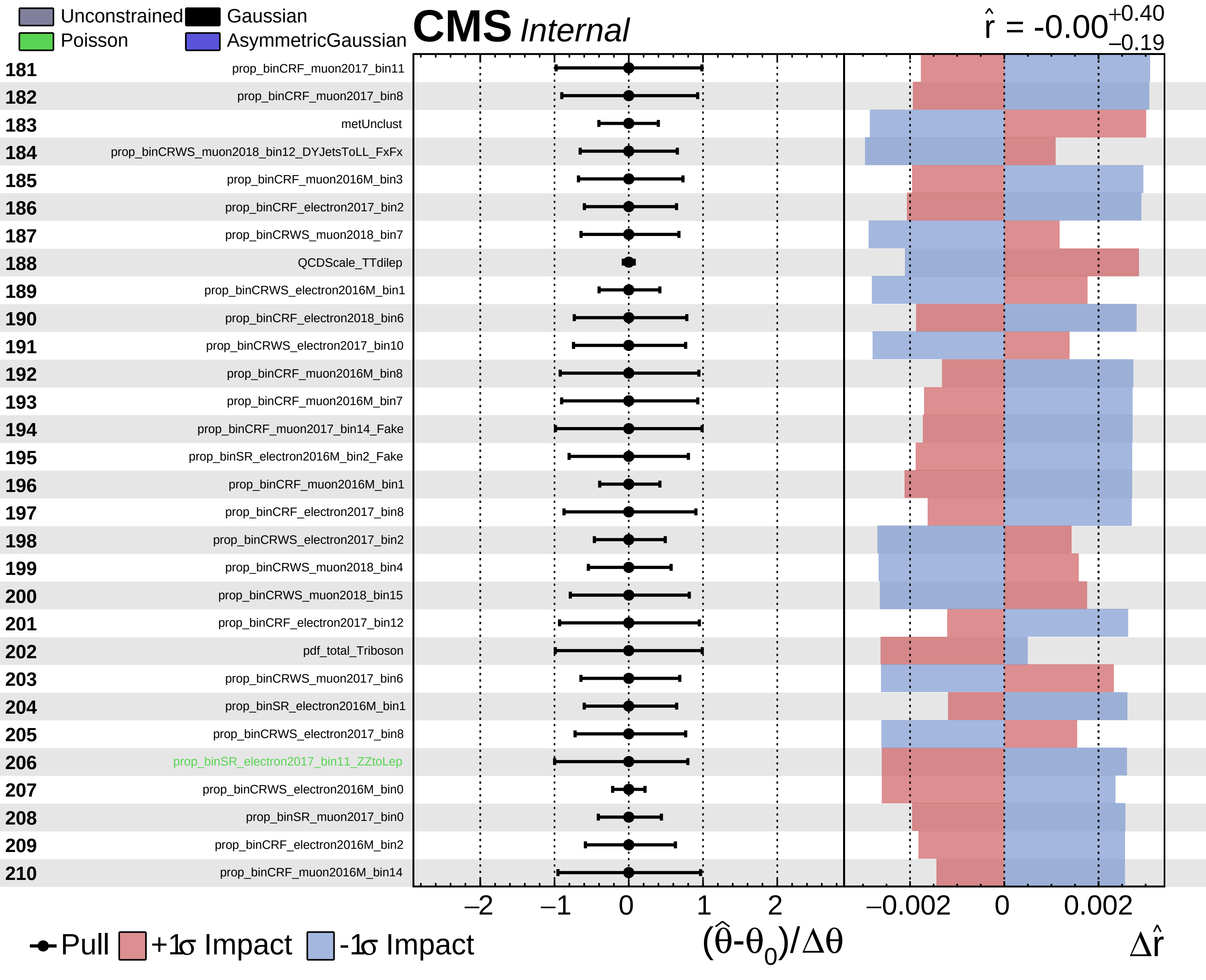
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.40}_{-0.19}$
91 prop_binCRWS_electron2018_bin5
92 prop_binSR_electron2018_bin6
93 prop_binSR_muon2016M_bin2
94 pdf_total_WpWpJJ_QCD
95 prop_binCRF_muon2017_bin3
96 prop_binCRF_muon2018_bin6
97 prop_binCRWS_muon2017_bin4
98 prop_binCRF_electron2018_bin0
99 prop_binCRWS_electron2018_bin9
100 prop_binCRF_muon2018_bin9
101 prop_binSR_electron2018_bin3
102 prop_binSR_electron2017_bin8
103 prop_binSR_electron2016M_bin11
104 PF
105 prop_binCRF_muon2017_bin2
106 prop_binCRWS_muon2018_bin3
107 prop_binSR_electron2017_bin6
108 prop_binCRF_electron2017_bin15_TTTo2L2Nu
109 prop_binCRTT_muon2018_bin0
110 prop_binCRF_muon2018_bin13
111 prop_binCRWS_muon2018_bin6
112 prop_binSR_muon2016M_bin5
113 prop_binCRF_muon2018_bin8
114 prop_binSR_muon2018_bin1
115 prop_binCRF_muon2018_bin14
116 prop_binSR_muon2017_bin1
117 prop_binCRF_muon2018_bin15
118 prop_binCRWS_electron2018_bin3
119 prop_binSR_electron2016M_bin8
120 prop_binSR_electron2017_bin7
−2 −1 0 1 2
−0.005 0 0.005
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



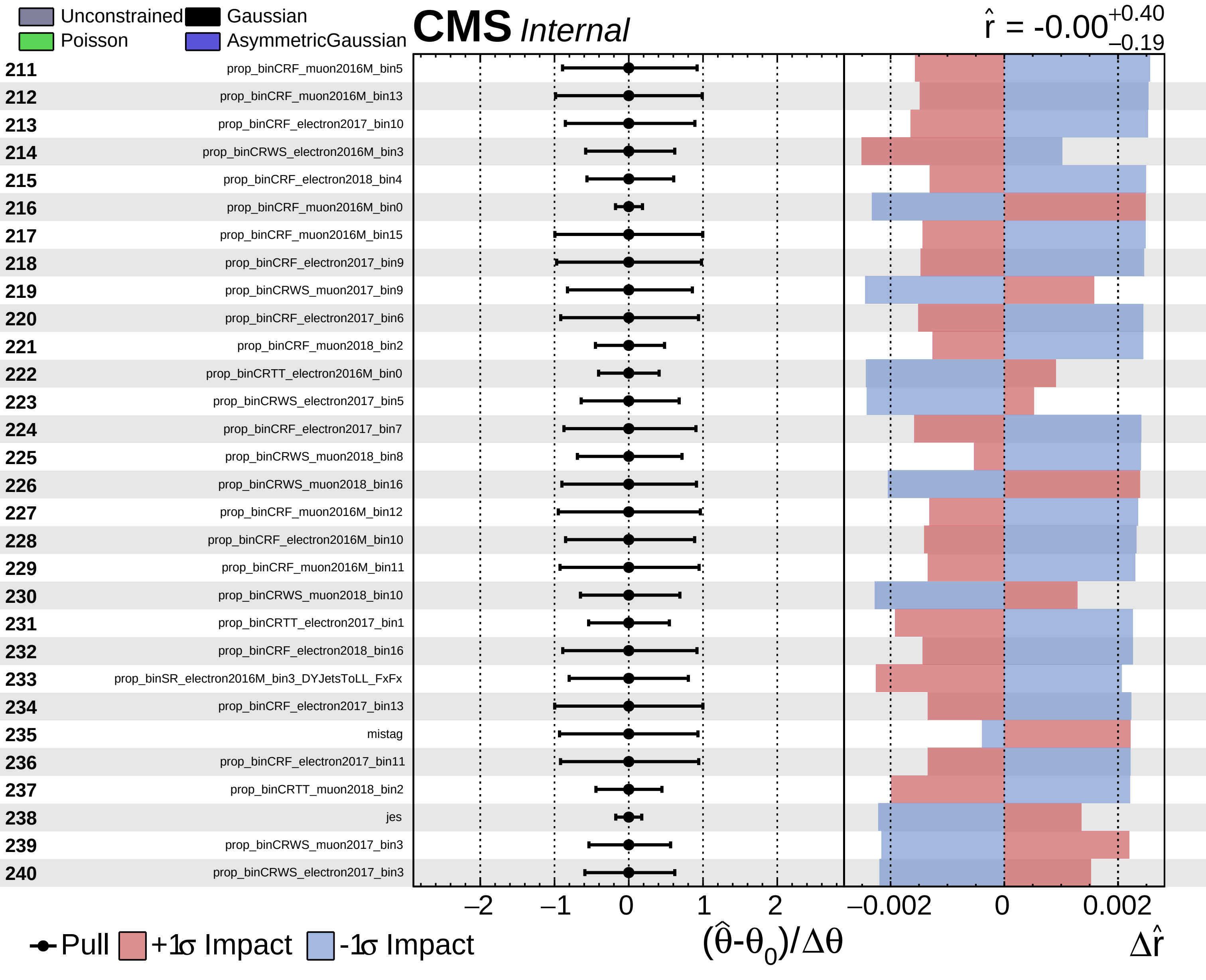
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.40}_{-0.19}$
121 prop_binCRF_electron2016M_bin12_DYJetsToLL_FxFx
122 prop_binCRF_muon2018_bin12
123 prop_binCRF_muon2018_bin5
124 btag
125 prop_binCRF_muon2017_bin4
126 prop_binCRWS_muon2016M_bin9
127 prop_binSR_electron2017_bin5
128 pdf_total_DY
129 prop_binCRWS_muon2017_bin1
130 prop_binCRF_muon2018_bin7
131 prop_binSR_electron2016M_bin6
132 prop_binCRF_electron2017_bin14_TVX
133 prop_binCRWS_electron2017_bin0
134 prop_binCRF_muon2018_bin11
135 prop_binCRF_muon2017_bin10
136 prop_binCRF_electron2018_bin10
137 prop_binSR_muon2017_bin5
138 prop_binCRF_electron2018_bin3
139 prop_binSR_muon2017_bin6_DYJetsToLL_FxFx
140 prop_binSR_muon2016M_bin1
141 prop_binSR_muon2017_bin6_Fake
142 prop_binCRF_electron2018_bin8
143 prop_binSR_electron2017_bin2
144 prop_binCRWS_muon2016M_bin1
145 prop_binCRF_muon2018_bin0
146 prop_binCRF_muon2016M_bin4
147 prop_binCRF_muon2017_bin5
148 pdf_total_TVX
149 prop_binSR_electron2017_bin3
150 prop_binSR_electron2016M_bin7
−2
−1
0
1
2
−0.005
0
0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



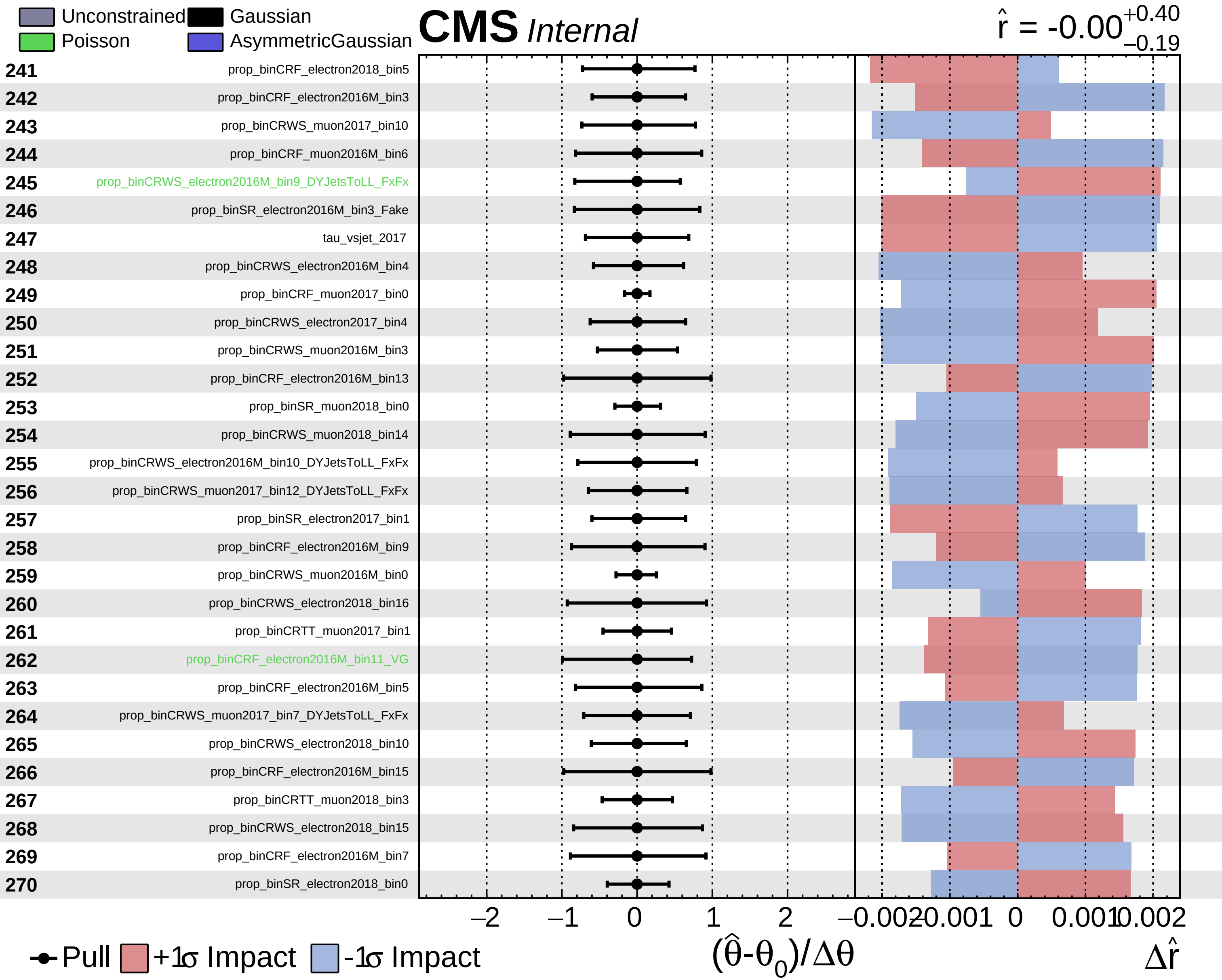
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.40}_{-0.19}$
151 prop_binCRWS_electron2018_bin7
152 prop_binSR_electron2018_bin2
153 prop_binCRF_electron2018_bin9
154 prop_binCRF_electron2017_bin3
155 prop_binCRF_muon2017_bin9
156 prop_binCRF_muon2016M_bin2
157 prop_binCRF_electron2016M_bin0
158 prop_binCRF_electron2018_bin11
159 prop_binSR_electron2016M_bin5
160 prop_binCRF_muon2016M_bin10
161 prop_binSR_electron2016M_bin4
162 prop_binCRF_muon2017_bin13
163 prop_binCRF_muon2017_bin6
164 prop_binCRF_electron2016M_bin4
165 prop_binCRWS_muon2018_bin1
166 prop_binCRF_electron2018_bin7
167 prop_binCRWS_muon2016M_bin7
168 prop_binCRF_electron2017_bin0
169 prop_binCRF_electron2018_bin13
170 prop_binCRF_electron2017_bin4
171 prop_binCRF_electron2018_bin2
172 prop_binCRTT_electron2018_bin2
173 prop_binCRTT_electron2018_bin0
174 prop_binCRF_electron2018_bin14
175 prop_binCRF_muon2017_bin7
176 prop_binCRWS_muon2018_bin2
177 prop_binCRF_muon2016M_bin9
178 prop_binCRF_muon2017_bin12
179 prop_binCRF_electron2018_bin12
180 prop_binCRF_electron2018_bin15
−2 −1 0 1 2
−0.004 −0.002 0 0.002 0.004
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



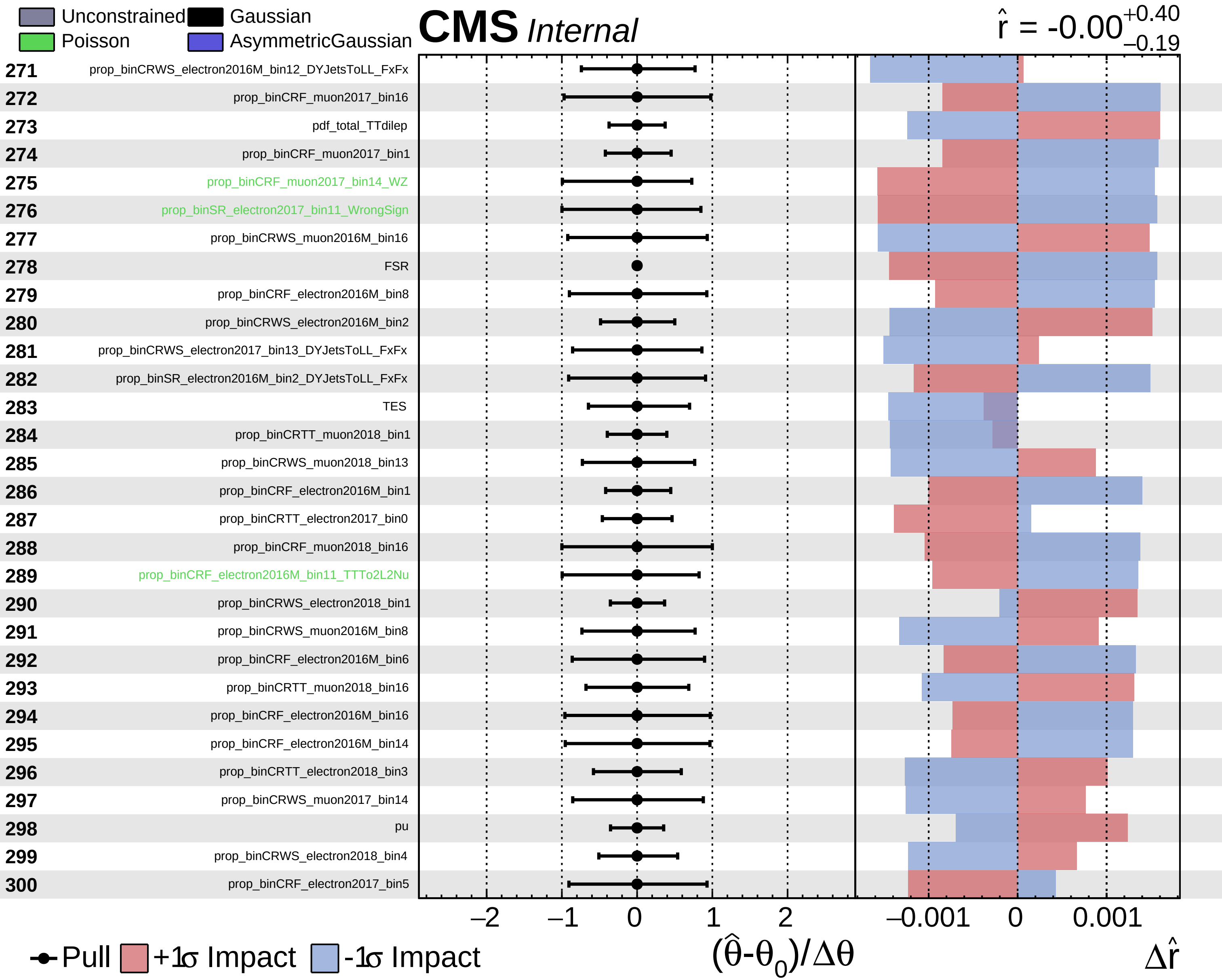
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.40}_{-0.19}$
181 prop_binCRF_muon2017_bin11
182 prop_binCRF_muon2017_bin8
183 metUnclust
184 prop_binCRWS_muon2018_bin12_DYJetsToLL_FxFx
185 prop_binCRF_muon2016M_bin3
186 prop_binCRF_electron2017_bin2
187 prop_binCRWS_muon2018_bin7
188 QCDScale_TTdilep
189 prop_binCRWS_electron2016M_bin1
190 prop_binCRF_electron2018_bin6
191 prop_binCRWS_electron2017_bin10
192 prop_binCRF_muon2016M_bin8
193 prop_binCRF_muon2016M_bin7
194 prop_binCRF_muon2017_bin14_Fake
195 prop_binSR_electron2016M_bin2_Fake
196 prop_binCRF_muon2016M_bin1
197 prop_binCRF_electron2017_bin8
198 prop_binCRWS_electron2017_bin2
199 prop_binCRWS_muon2018_bin4
200 prop_binCRWS_muon2018_bin15
201 prop_binCRF_electron2017_bin12
202 pdf_total_Triboson
203 prop_binCRWS_muon2017_bin6
204 prop_binSR_electron2016M_bin1
205 prop_binCRWS_electron2017_bin8
206 prop_binSR_electron2017_bin11_ZZtoLep
207 prop_binCRWS_electron2016M_bin0
208 prop_binSR_muon2017_bin0
209 prop_binCRF_electron2016M_bin2
210 prop_binCRF_muon2016M_bin14
−2
−1
0
1
2
−0.002
0
0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



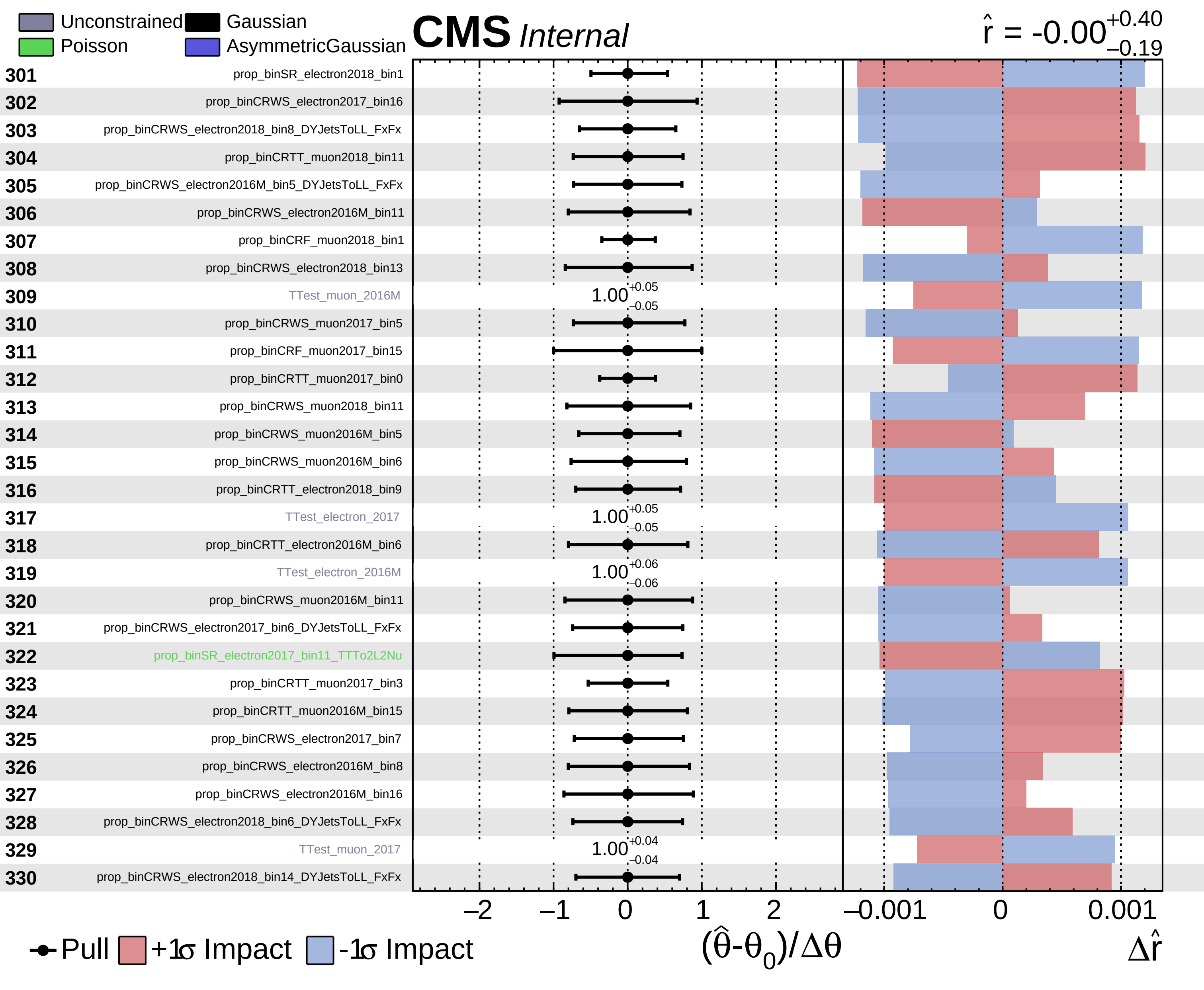
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = -0.00 +0.40 −0.19
211 prop_binCRF_muon2016M_bin5
212 prop_binCRF_muon2016M_bin13
213 prop_binCRF_electron2017_bin10
214 prop_binCRWS_electron2016M_bin3
215 prop_binCRF_electron2018_bin4
216 prop_binCRF_muon2016M_bin0
217 prop_binCRF_muon2016M_bin15
218 prop_binCRF_electron2017_bin9
219 prop_binCRWS_muon2017_bin9
220 prop_binCRF_electron2017_bin6
221 prop_binCRF_muon2018_bin2
222 prop_binCRTT_electron2016M_bin0
223 prop_binCRWS_electron2017_bin5
224 prop_binCRF_electron2017_bin7
225 prop_binCRWS_muon2018_bin8
226 prop_binCRWS_muon2018_bin16
227 prop_binCRF_muon2016M_bin12
228 prop_binCRF_electron2016M_bin10
229 prop_binCRF_muon2016M_bin11
230 prop_binCRWS_muon2018_bin10
231 prop_binCRTT_electron2017_bin1
232 prop_binCRF_electron2018_bin16
233 prop_binSR_electron2016M_bin3_DYJetsToLL_FxFx
234 prop_binCRF_electron2017_bin13
235 mistag
236 prop_binCRF_electron2017_bin11
237 prop_binCRTT_muon2018_bin2
238 jes
239 prop_binCRWS_muon2017_bin3
240 prop_binCRWS_electron2017_bin3
−2 −1 0 1 2
−0.002 0 0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

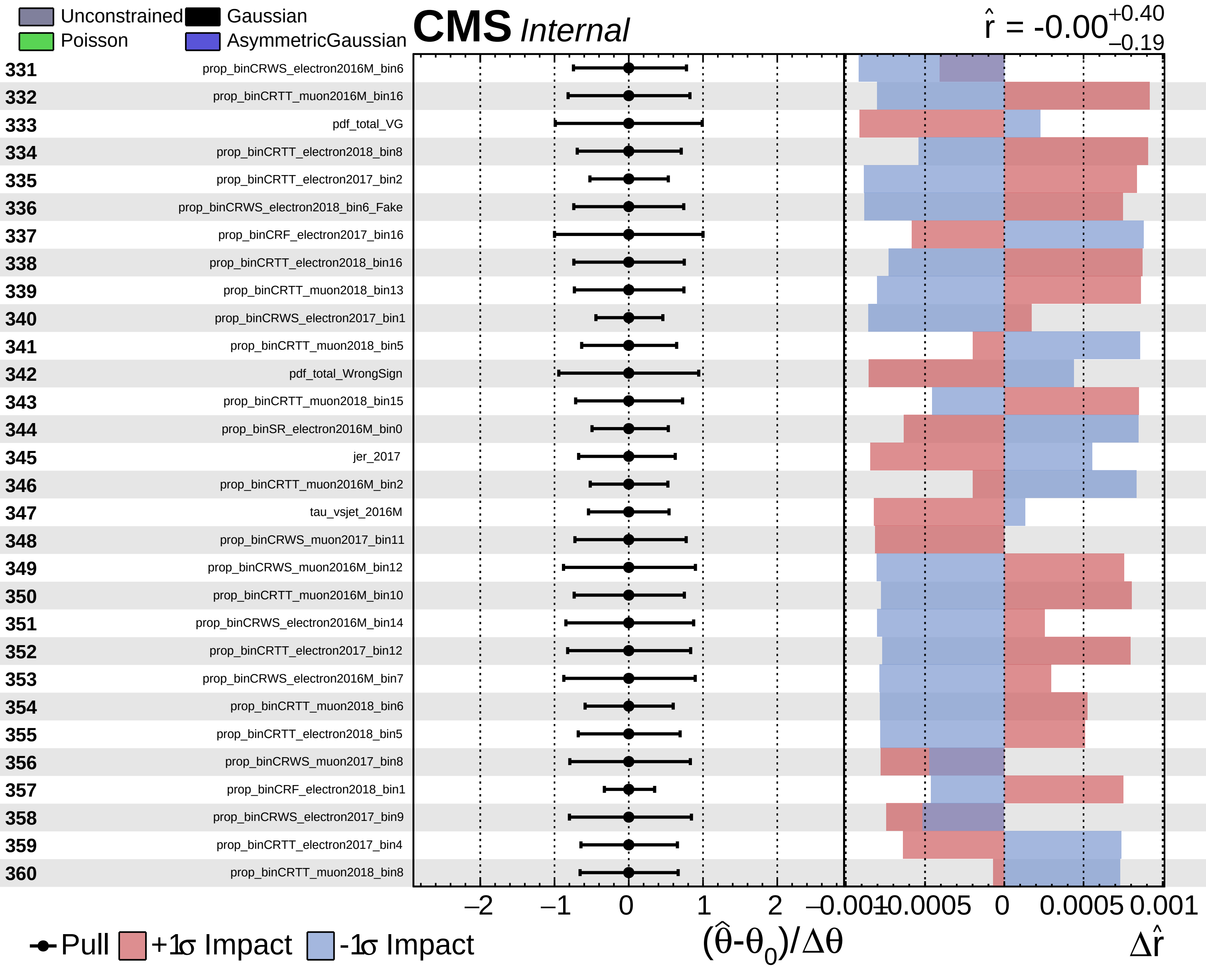


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.40}_{-0.19}$
241 prop_binCRF_electron2018_bin5
242 prop_binCRF_electron2016M_bin3
243 prop_binCRWS_muon2017_bin10
244 prop_binCRF_muon2016M_bin6
245 prop_binCRWS_electron2016M_bin9_DYJetsToLL_FxFx
246 prop_binSR_electron2016M_bin3_Fake
247 tau_vsjet_2017
248 prop_binCRWS_electron2016M_bin4
249 prop_binCRF_muon2017_bin0
250 prop_binCRWS_electron2017_bin4
251 prop_binCRWS_muon2016M_bin3
252 prop_binCRF_electron2016M_bin13
253 prop_binSR_muon2018_bin0
254 prop_binCRWS_muon2018_bin14
255 prop_binCRWS_electron2016M_bin10_DYJetsToLL_FxFx
256 prop_binCRWS_muon2017_bin12_DYJetsToLL_FxFx
257 prop_binSR_electron2017_bin1
258 prop_binCRF_electron2016M_bin9
259 prop_binCRWS_muon2016M_bin0
260 prop_binCRWS_electron2018_bin16
261 prop_binCRTT_muon2017_bin1
262 prop_binCRF_electron2016M_bin11_VG
263 prop_binCRF_electron2016M_bin5
264 prop_binCRWS_muon2017_bin7_DYJetsToLL_FxFx
265 prop_binCRWS_electron2018_bin10
266 prop_binCRF_electron2016M_bin15
267 prop_binCRTT_muon2018_bin3
268 prop_binCRWS_electron2018_bin15
269 prop_binCRF_electron2016M_bin7
270 prop_binSR_electron2018_bin0
−2 −1 0 1 2
−0.002 −0.001 0 0.001 0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

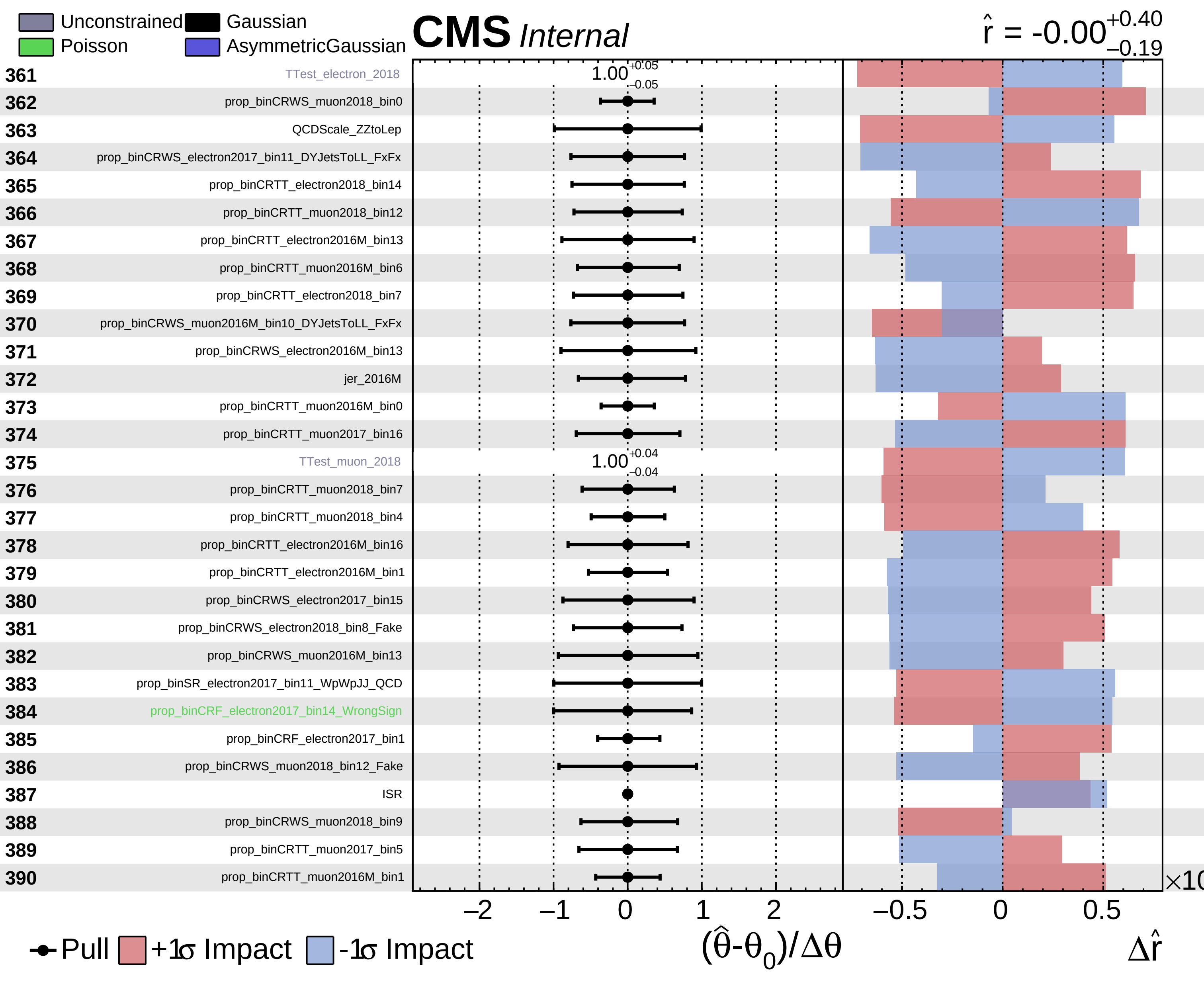


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.40}_{-0.19}$
271 prop_binCRWS_electron2016M_bin12_DYJetsToLL_FxFx
272 prop_binCRF_muon2017_bin16
273 pdf_total_TTdilep
274 prop_binCRF_muon2017_bin1
275 prop_binCRF_muon2017_bin14_WZ
276 prop_binSR_electron2017_bin11_WrongSign
277 prop_binCRWS_muon2016M_bin16
278 FSR
279 prop_binCRF_electron2016M_bin8
280 prop_binCRWS_electron2016M_bin2
281 prop_binCRWS_electron2017_bin13_DYJetsToLL_FxFx
282 prop_binSR_electron2016M_bin2_DYJetsToLL_FxFx
283 TES
284 prop_binCRTT_muon2018_bin1
285 prop_binCRWS_muon2018_bin13
286 prop_binCRF_electron2016M_bin1
287 prop_binCRTT_electron2017_bin0
288 prop_binCRF_muon2018_bin16
289 prop_binCRF_electron2016M_bin11_TTTo2L2Nu
290 prop_binCRWS_electron2018_bin1
291 prop_binCRWS_muon2016M_bin8
292 prop_binCRF_electron2016M_bin6
293 prop_binCRTT_muon2018_bin16
294 prop_binCRF_electron2016M_bin16
295 prop_binCRF_electron2016M_bin14
296 prop_binCRTT_electron2018_bin3
297 prop_binCRWS_muon2017_bin14
298 pu
299 prop_binCRWS_electron2018_bin4
300 prop_binCRF_electron2017_bin5
−2
−1
0
1
2
−0.001
0
0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

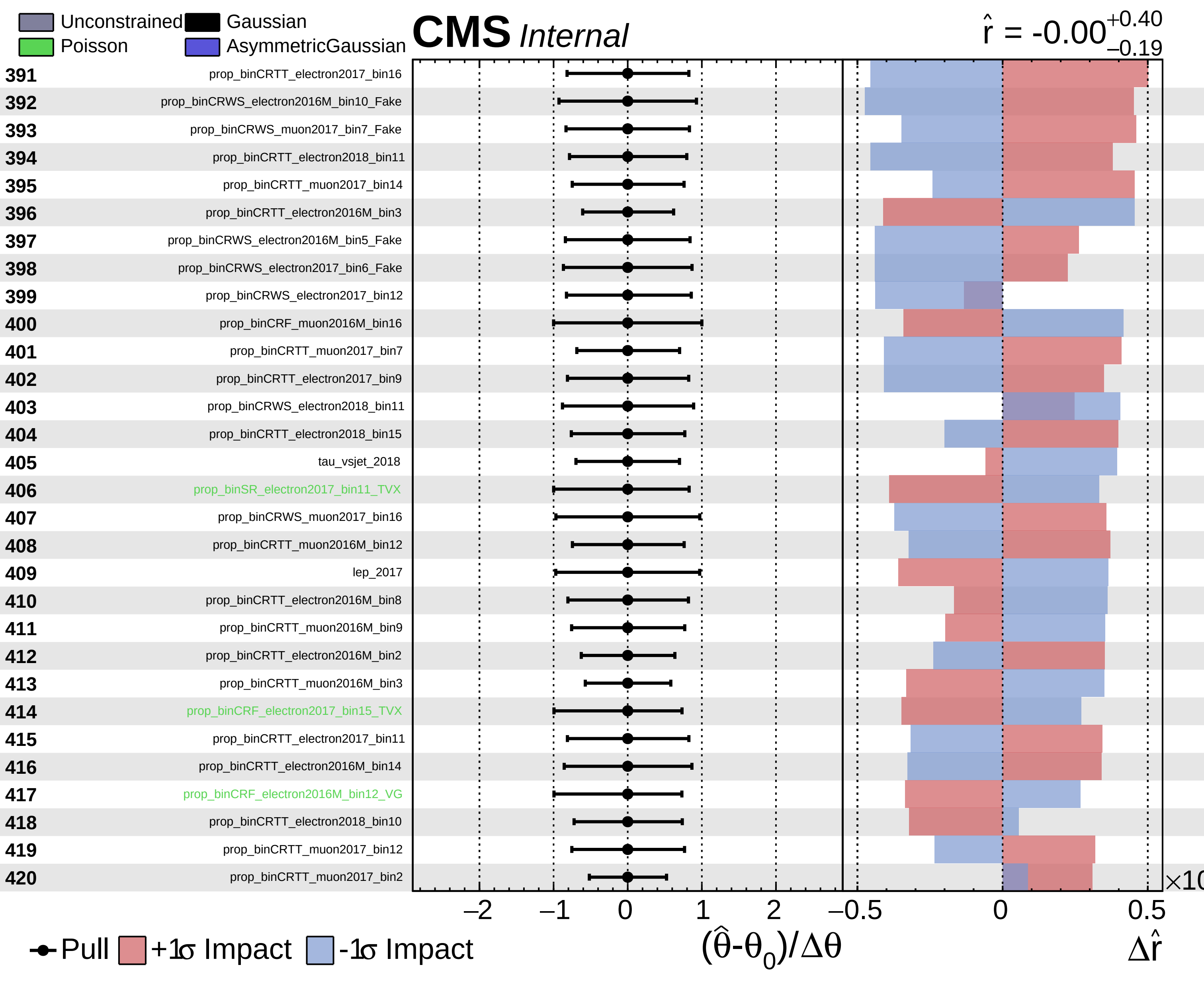

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.40}_{-0.19}$
301 prop_binSR_electron2018_bin1
302 prop_binCRWS_electron2017_bin16
303 prop_binCRWS_electron2018_bin8_DYJetsToLL_FxFx
304 prop_binCRTT_muon2018_bin11
305 prop_binCRWS_electron2016M_bin5_DYJetsToLL_FxFx
306 prop_binCRWS_electron2016M_bin11
307 prop_binCRF_muon2018_bin1
308 prop_binCRWS_electron2018_bin13
309 TTest_muon_2016M $1.00^{+0.05}_{-0.05}$
310 prop_binCRWS_muon2017_bin5
311 prop_binCRF_muon2017_bin15
312 prop_binCRTT_muon2017_bin0
313 prop_binCRWS_muon2018_bin11
314 prop_binCRWS_muon2016M_bin5
315 prop_binCRWS_muon2016M_bin6
316 prop_binCRTT_electron2018_bin9
317 TTest_electron_2017 $1.00^{+0.05}_{-0.05}$
318 prop_binCRTT_electron2016M_bin6
319 TTest_electron_2016M $1.00^{+0.06}_{-0.06}$
320 prop_binCRWS_muon2016M_bin11
321 prop_binCRWS_electron2017_bin6_DYJetsToLL_FxFx
322 prop_binSR_electron2017_bin11_TTTo2L2Nu
323 prop_binCRTT_muon2017_bin3
324 prop_binCRTT_muon2016M_bin15
325 prop_binCRWS_electron2017_bin7
326 prop_binCRWS_electron2016M_bin8
327 prop_binCRWS_electron2016M_bin16
328 prop_binCRWS_electron2018_bin6_DYJetsToLL_FxFx
329 TTest_muon_2017 $1.00^{+0.04}_{-0.04}$
330 prop_binCRWS_electron2018_bin14_DYJetsToLL_FxFx
−2 −1 0 1 2
−0.001 0 0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



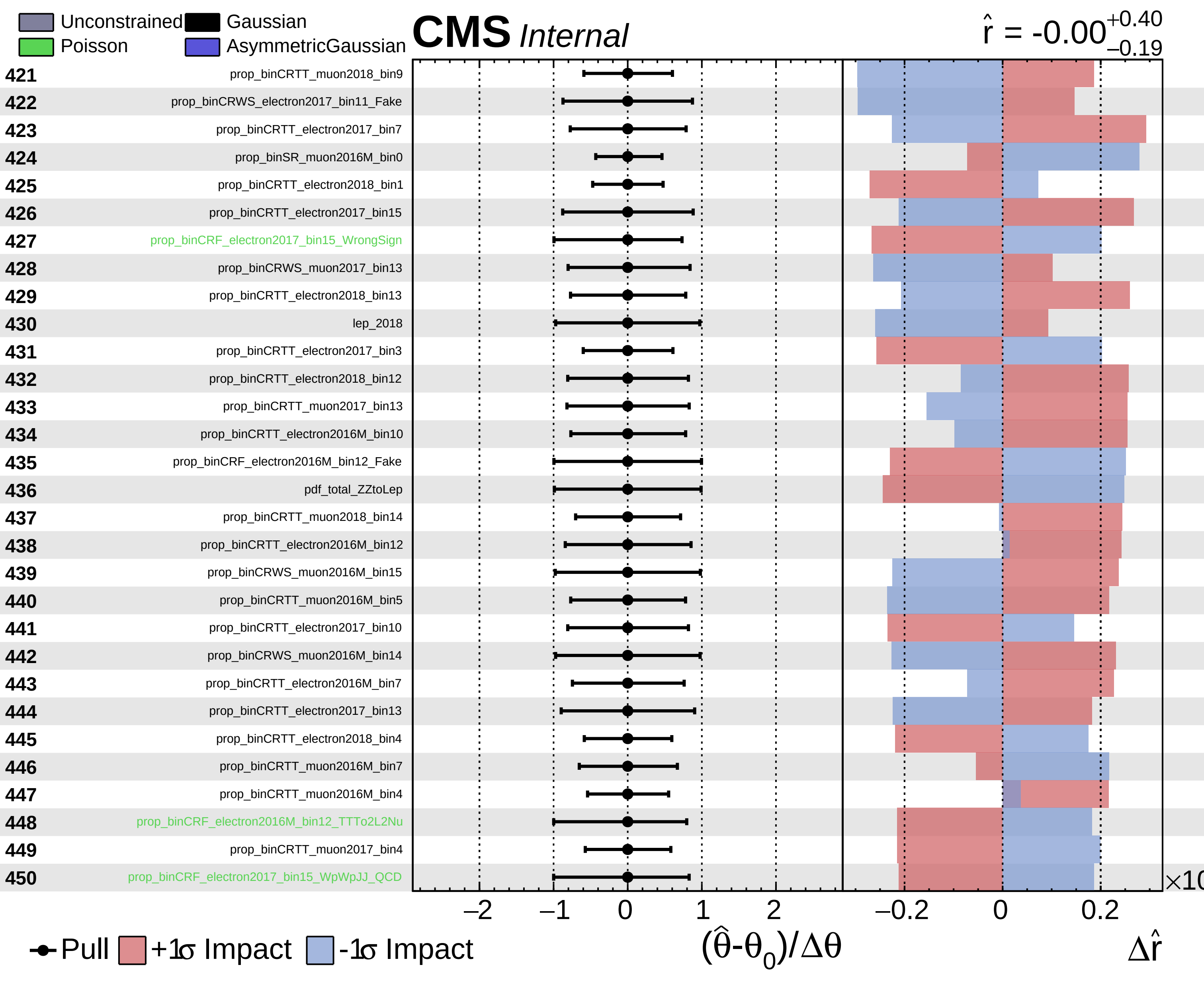
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.40}_{-0.19}$
331 prop_binCRWS_electron2016M_bin6
332 prop_binCRTT_muon2016M_bin16
333 pdf_total_VG
334 prop_binCRTT_electron2018_bin8
335 prop_binCRTT_electron2017_bin2
336 prop_binCRWS_electron2018_bin6_Fake
337 prop_binCRF_electron2017_bin16
338 prop_binCRTT_electron2018_bin16
339 prop_binCRTT_muon2018_bin13
340 prop_binCRWS_electron2017_bin1
341 prop_binCRTT_muon2018_bin5
342 pdf_total_WrongSign
343 prop_binCRTT_muon2018_bin15
344 prop_binSR_electron2016M_bin0
345 jer_2017
346 prop_binCRTT_muon2016M_bin2
347 tau_vsjet_2016M
348 prop_binCRWS_muon2017_bin11
349 prop_binCRWS_muon2016M_bin12
350 prop_binCRTT_muon2016M_bin10
351 prop_binCRWS_electron2016M_bin14
352 prop_binCRTT_electron2017_bin12
353 prop_binCRWS_electron2016M_bin7
354 prop_binCRTT_muon2018_bin6
355 prop_binCRTT_electron2018_bin5
356 prop_binCRWS_muon2017_bin8
357 prop_binCRF_electron2018_bin1
358 prop_binCRWS_electron2017_bin9
359 prop_binCRTT_electron2017_bin4
360 prop_binCRTT_muon2018_bin8
−2 −1 0 1 2
−0.001 −0.0005 0 0.0005 0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



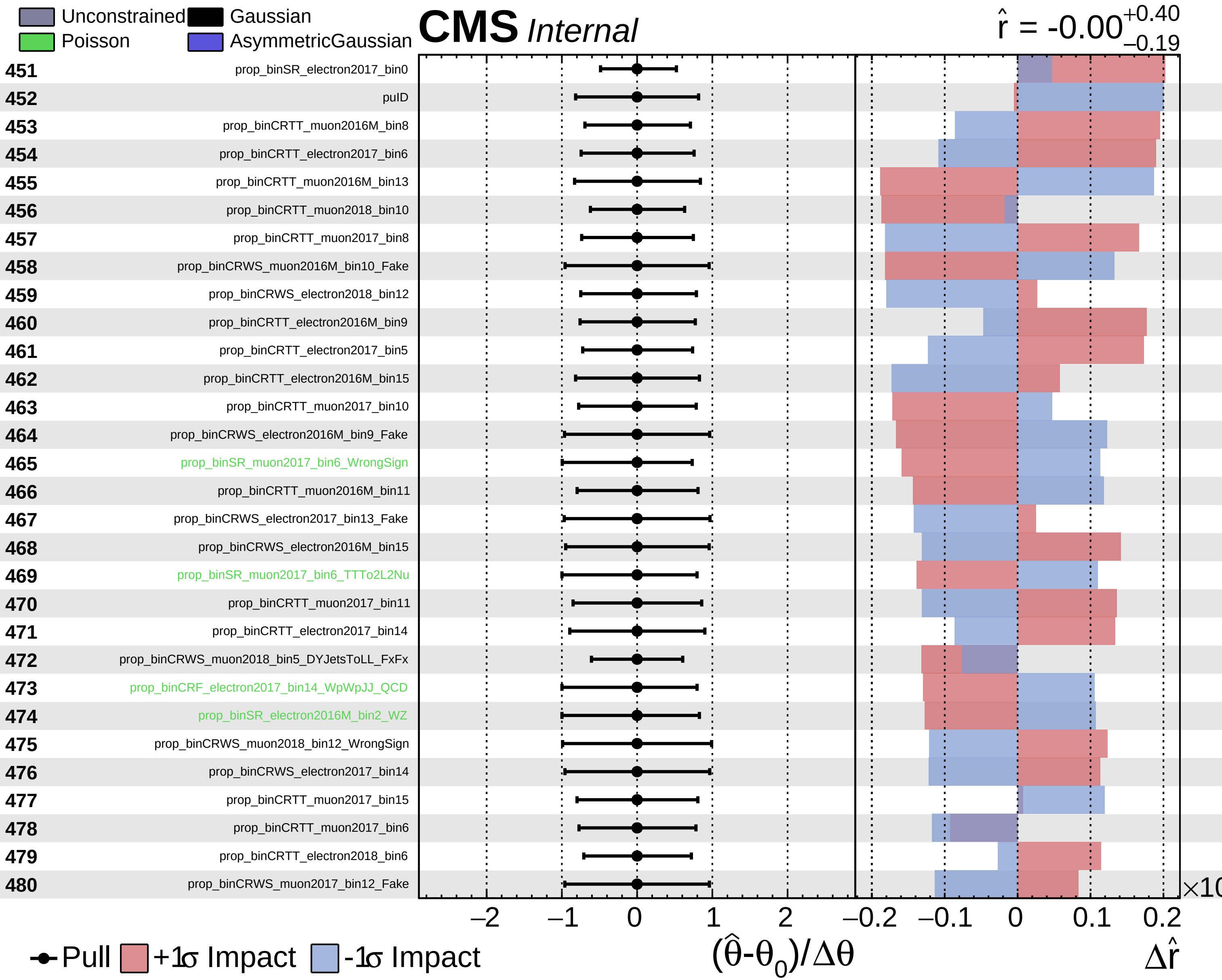
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.40}_{-0.19}$
361 TTest_electron_2018 $1.00^{+0.05}_{-0.05}$
362 prop_binCRWS_muon2018_bin0
363 QCDScale_ZZtoLep
364 prop_binCRWS_electron2017_bin11_DYJetsToLL_FxFx
365 prop_binCRTT_electron2018_bin14
366 prop_binCRTT_muon2018_bin12
367 prop_binCRTT_electron2016M_bin13
368 prop_binCRTT_muon2016M_bin6
369 prop_binCRTT_electron2018_bin7
370 prop_binCRWS_muon2016M_bin10_DYJetsToLL_FxFx
371 prop_binCRWS_electron2016M_bin13
372 jer_2016M
373 prop_binCRTT_muon2016M_bin0
374 prop_binCRTT_muon2017_bin16
375 TTest_muon_2018 $1.00^{+0.04}_{-0.04}$
376 prop_binCRTT_muon2018_bin7
377 prop_binCRTT_muon2018_bin4
378 prop_binCRTT_electron2016M_bin16
379 prop_binCRTT_electron2016M_bin1
380 prop_binCRWS_electron2017_bin15
381 prop_binCRWS_electron2018_bin8_Fake
382 prop_binCRWS_muon2016M_bin13
383 prop_binSR_electron2017_bin11_WpWpJJ_QCD
384 prop_binCRF_electron2017_bin14_WrongSign
385 prop_binCRF_electron2017_bin1
386 prop_binCRWS_muon2018_bin12_Fake
387 ISR
388 prop_binCRWS_muon2018_bin9
389 prop_binCRTT_muon2017_bin5
390 prop_binCRTT_muon2016M_bin1
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



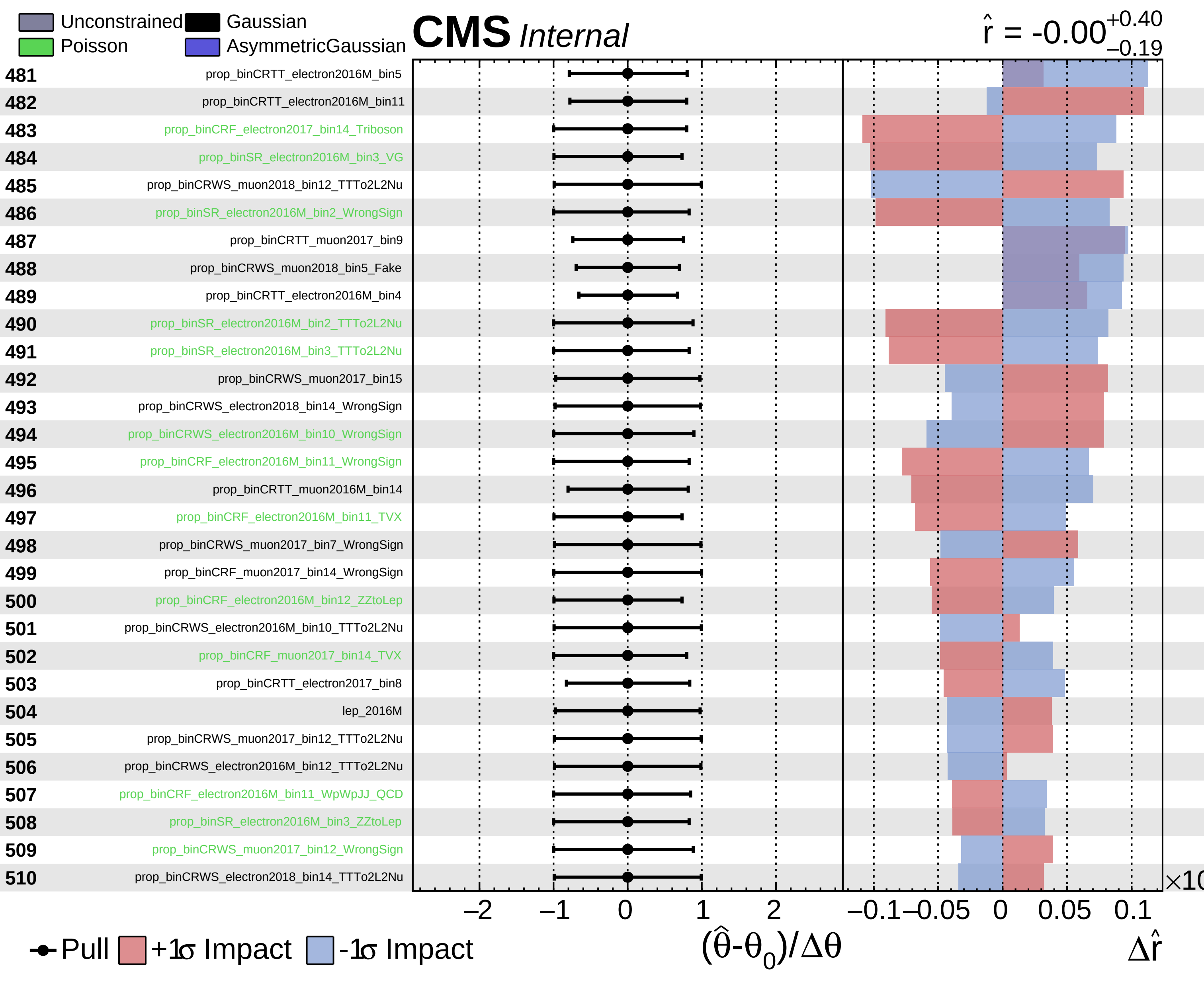
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.40}_{-0.19}$
391 prop_binCRTT_electron2017_bin16
392 prop_binCRWS_electron2016M_bin10_Fake
393 prop_binCRWS_muon2017_bin7_Fake
394 prop_binCRTT_electron2018_bin11
395 prop_binCRTT_muon2017_bin14
396 prop_binCRTT_electron2016M_bin3
397 prop_binCRWS_electron2016M_bin5_Fake
398 prop_binCRWS_electron2017_bin6_Fake
399 prop_binCRWS_electron2017_bin12
400 prop_binCRF_muon2016M_bin16
401 prop_binCRTT_muon2017_bin7
402 prop_binCRTT_electron2017_bin9
403 prop_binCRWS_electron2018_bin11
404 prop_binCRTT_electron2018_bin15
405 tau_vsjet_2018
406 prop_binSR_electron2017_bin11_TVX
407 prop_binCRWS_muon2017_bin16
408 prop_binCRTT_muon2016M_bin12
409 lep_2017
410 prop_binCRTT_electron2016M_bin8
411 prop_binCRTT_muon2016M_bin9
412 prop_binCRTT_electron2016M_bin2
413 prop_binCRTT_muon2016M_bin3
414 prop_binCRF_electron2017_bin15_TVX
415 prop_binCRTT_electron2017_bin11
416 prop_binCRTT_electron2016M_bin14
417 prop_binCRF_electron2016M_bin12_VG
418 prop_binCRTT_electron2018_bin10
419 prop_binCRTT_muon2017_bin12
420 prop_binCRTT_muon2017_bin2
−2
−1
0
1
2
−0.5
0
0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

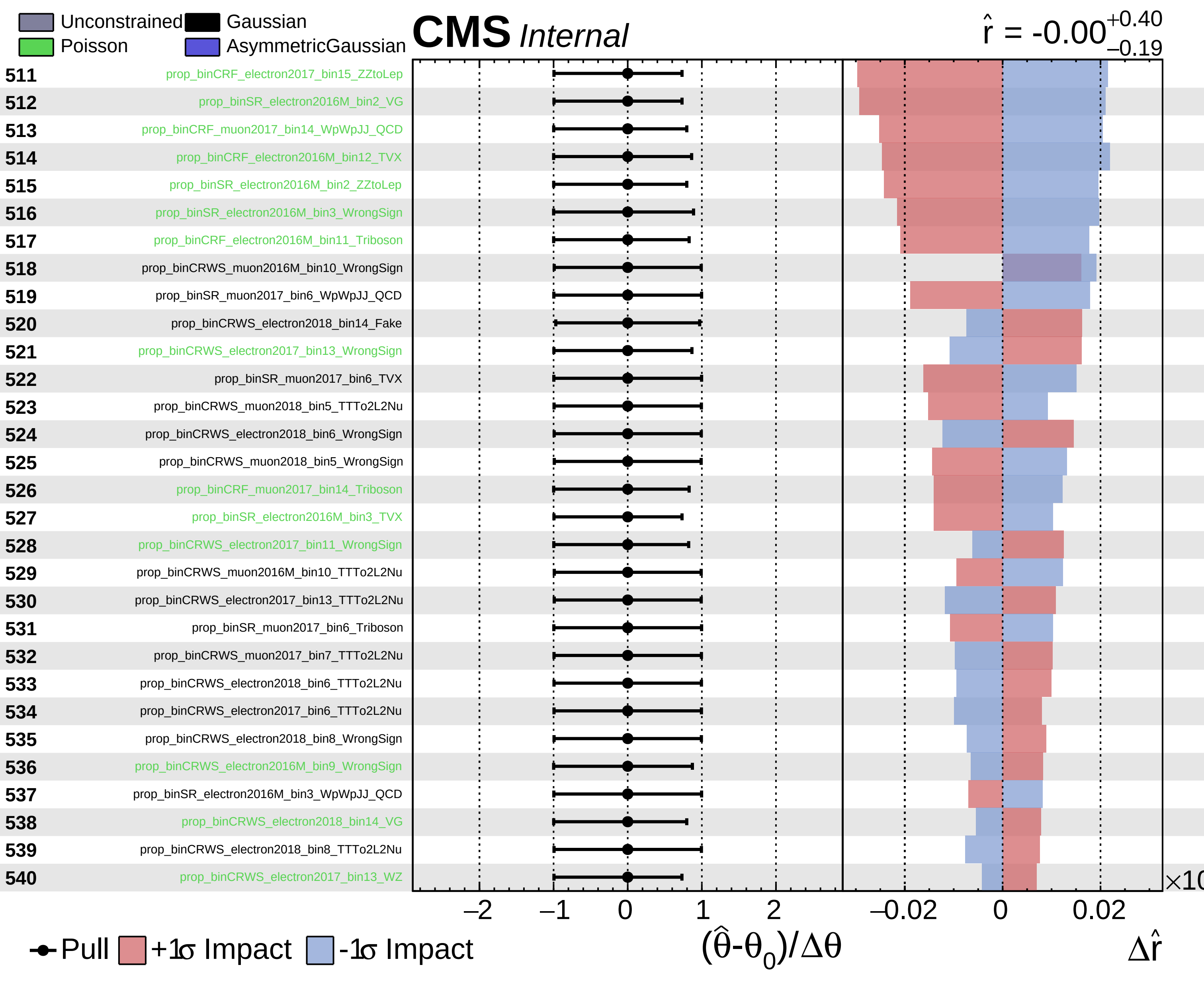


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.40}_{-0.19}$
421 prop_binCRTT_muon2018_bin9
422 prop_binCRWS_electron2017_bin11_Fake
423 prop_binCRTT_electron2017_bin7
424 prop_binSR_muon2016M_bin0
425 prop_binCRTT_electron2018_bin1
426 prop_binCRTT_electron2017_bin15
427 prop_binCRF_electron2017_bin15_WrongSign
428 prop_binCRWS_muon2017_bin13
429 prop_binCRTT_electron2018_bin13
430 lep_2018
431 prop_binCRTT_electron2017_bin3
432 prop_binCRTT_electron2018_bin12
433 prop_binCRTT_muon2017_bin13
434 prop_binCRTT_electron2016M_bin10
435 prop_binCRF_electron2016M_bin12_Fake
436 pdf_total_ZZtoLep
437 prop_binCRTT_muon2018_bin14
438 prop_binCRTT_electron2016M_bin12
439 prop_binCRWS_muon2016M_bin15
440 prop_binCRTT_muon2016M_bin5
441 prop_binCRTT_electron2017_bin10
442 prop_binCRWS_muon2016M_bin14
443 prop_binCRTT_electron2016M_bin7
444 prop_binCRTT_electron2017_bin13
445 prop_binCRTT_electron2018_bin4
446 prop_binCRTT_muon2016M_bin7
447 prop_binCRTT_muon2016M_bin4
448 prop_binCRF_electron2016M_bin12_TTTo2L2Nu
449 prop_binCRTT_muon2017_bin4
450 prop_binCRF_electron2017_bin15_WpWpJJ_QCD
−2 −1 0 1 2
−0.2 0 0.2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

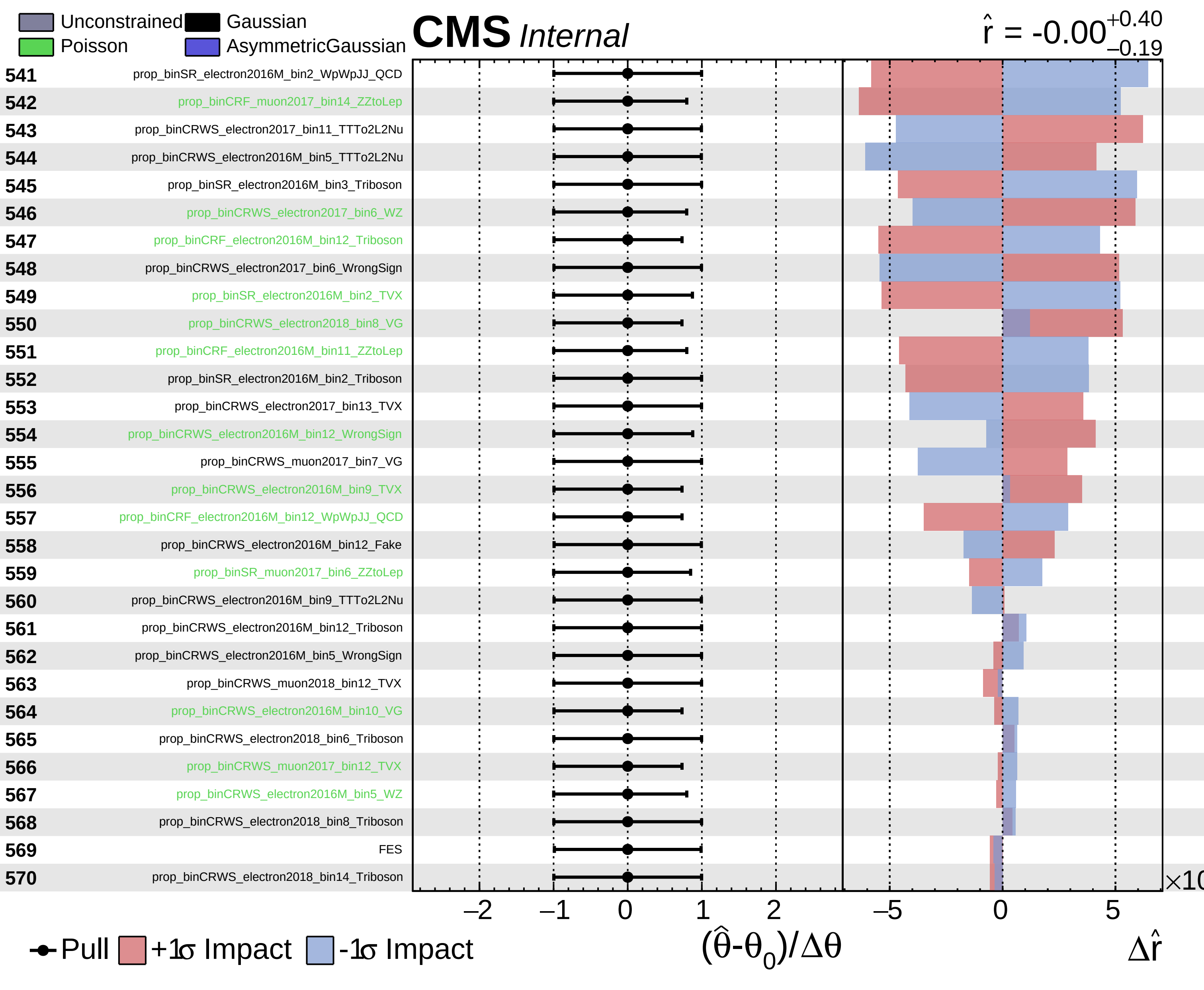


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.40}_{-0.19}$
451 prop_binSR_electron2017_bin0
452 puID
453 prop_binCRTT_muon2016M_bin8
454 prop_binCRTT_electron2017_bin6
455 prop_binCRTT_muon2016M_bin13
456 prop_binCRTT_muon2018_bin10
457 prop_binCRTT_muon2017_bin8
458 prop_binCRWS_muon2016M_bin10_Fake
459 prop_binCRWS_electron2018_bin12
460 prop_binCRTT_electron2016M_bin9
461 prop_binCRTT_electron2017_bin5
462 prop_binCRTT_electron2016M_bin15
463 prop_binCRTT_muon2017_bin10
464 prop_binCRWS_electron2016M_bin9_Fake
465 prop_binSR_muon2017_bin6_WrongSign
466 prop_binCRTT_muon2016M_bin11
467 prop_binCRWS_electron2017_bin13_Fake
468 prop_binCRWS_electron2016M_bin15
469 prop_binSR_muon2017_bin6_TTTo2L2Nu
470 prop_binCRTT_muon2017_bin11
471 prop_binCRTT_electron2017_bin14
472 prop_binCRWS_muon2018_bin5_DYJetsToLL_FxFx
473 prop_binCRF_electron2017_bin14_WpWpJJ_QCD
474 prop_binSR_electron2016M_bin2_WZ
475 prop_binCRWS_muon2018_bin12_WrongSign
476 prop_binCRWS_electron2017_bin14
477 prop_binCRTT_muon2017_bin15
478 prop_binCRTT_muon2017_bin6
479 prop_binCRTT_electron2018_bin6
480 prop_binCRWS_muon2017_bin12_Fake
−2 −1 0 1 2
−0.2 −0.1 0 0.1 0.2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

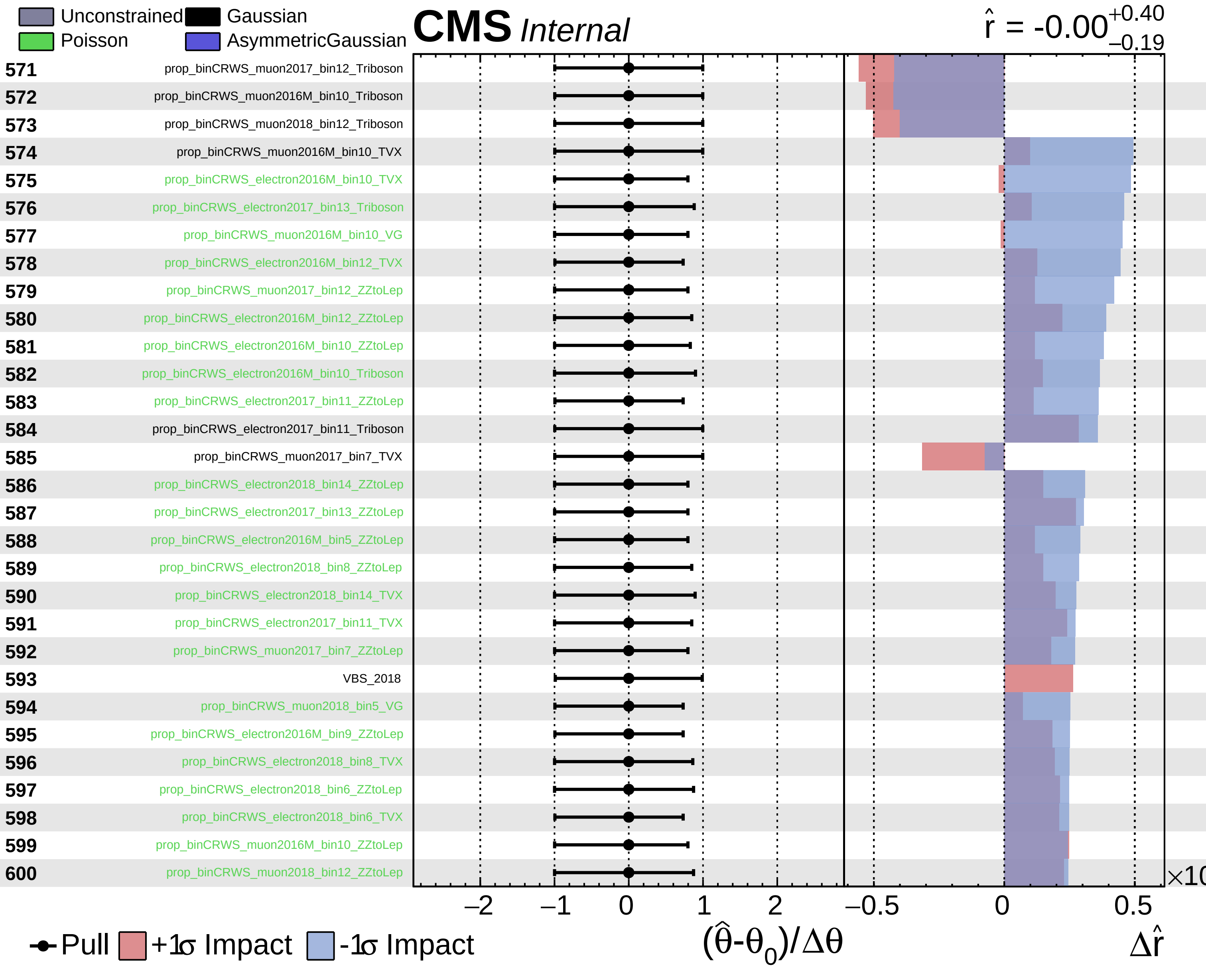

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.40}_{-0.19}$
481 prop_binCRTT_electron2016M_bin5
482 prop_binCRTT_electron2016M_bin11
483 prop_binCRF_electron2017_bin14_Triboson
484 prop_binSR_electron2016M_bin3_VG
485 prop_binCRWS_muon2018_bin12_TTTo2L2Nu
486 prop_binSR_electron2016M_bin2_WrongSign
487 prop_binCRTT_muon2017_bin9
488 prop_binCRWS_muon2018_bin5_Fake
489 prop_binCRTT_electron2016M_bin4
490 prop_binSR_electron2016M_bin2_TTTo2L2Nu
491 prop_binSR_electron2016M_bin3_TTTo2L2Nu
492 prop_binCRWS_muon2017_bin15
493 prop_binCRWS_electron2018_bin14_WrongSign
494 prop_binCRWS_electron2016M_bin10_WrongSign
495 prop_binCRF_electron2016M_bin11_WrongSign
496 prop_binCRTT_muon2016M_bin14
497 prop_binCRF_electron2016M_bin11_TVX
498 prop_binCRWS_muon2017_bin7_WrongSign
499 prop_binCRF_muon2017_bin14_WrongSign
500 prop_binCRF_electron2016M_bin12_ZZtoLep
501 prop_binCRWS_electron2016M_bin10_TTTo2L2Nu
502 prop_binCRF_muon2017_bin14_TVX
503 prop_binCRTT_electron2017_bin8
504 lep_2016M
505 prop_binCRWS_muon2017_bin12_TTTo2L2Nu
506 prop_binCRWS_electron2016M_bin12_TTTo2L2Nu
507 prop_binCRF_electron2016M_bin11_WpWpJJ_QCD
508 prop_binSR_electron2016M_bin3_ZZtoLep
509 prop_binCRWS_muon2017_bin12_WrongSign
510 prop_binCRWS_electron2018_bin14_TTTo2L2Nu
−2 −1 0 1 2
−0.1 −0.05 0 0.05 0.1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



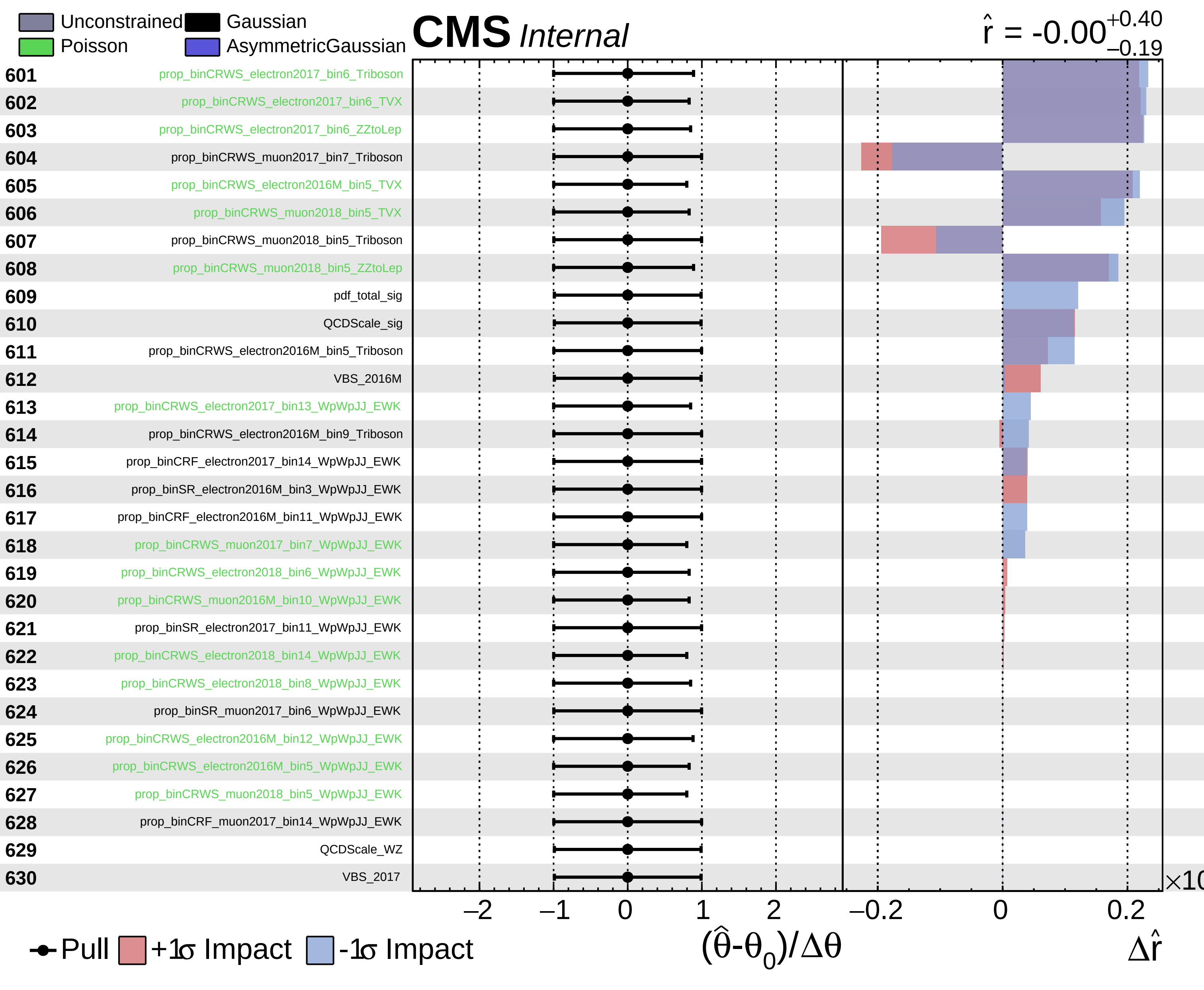
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.40}_{-0.19}$
511 prop_binCRF_electron2017_bin15_ZZtoLep
512 prop_binSR_electron2016M_bin2_VG
513 prop_binCRF_muon2017_bin14_WpWpJJ_QCD
514 prop_binCRF_electron2016M_bin12_TVX
515 prop_binSR_electron2016M_bin2_ZZtoLep
516 prop_binSR_electron2016M_bin3_WrongSign
517 prop_binCRF_electron2016M_bin11_Triboson
518 prop_binCRWS_muon2016M_bin10_WrongSign
519 prop_binSR_muon2017_bin6_WpWpJJ_QCD
520 prop_binCRWS_electron2018_bin14_Fake
521 prop_binCRWS_electron2017_bin13_WrongSign
522 prop_binSR_muon2017_bin6_TVX
523 prop_binCRWS_muon2018_bin5_TTTo2L2Nu
524 prop_binCRWS_electron2018_bin6_WrongSign
525 prop_binCRWS_muon2018_bin5_WrongSign
526 prop_binCRF_muon2017_bin14_Triboson
527 prop_binSR_electron2016M_bin3_TVX
528 prop_binCRWS_electron2017_bin11_WrongSign
529 prop_binCRWS_muon2016M_bin10_TTTo2L2Nu
530 prop_binCRWS_electron2017_bin13_TTTo2L2Nu
531 prop_binSR_muon2017_bin6_Triboson
532 prop_binCRWS_muon2017_bin7_TTTo2L2Nu
533 prop_binCRWS_electron2018_bin6_TTTo2L2Nu
534 prop_binCRWS_electron2017_bin6_TTTo2L2Nu
535 prop_binCRWS_electron2018_bin8_WrongSign
536 prop_binCRWS_electron2016M_bin9_WrongSign
537 prop_binSR_electron2016M_bin3_WpWpJJ_QCD
538 prop_binCRWS_electron2018_bin14_VG
539 prop_binCRWS_electron2018_bin8_TTTo2L2Nu
540 prop_binCRWS_electron2017_bin13_WZ
−2 −1 0 1 2
−0.02 0 0.02
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



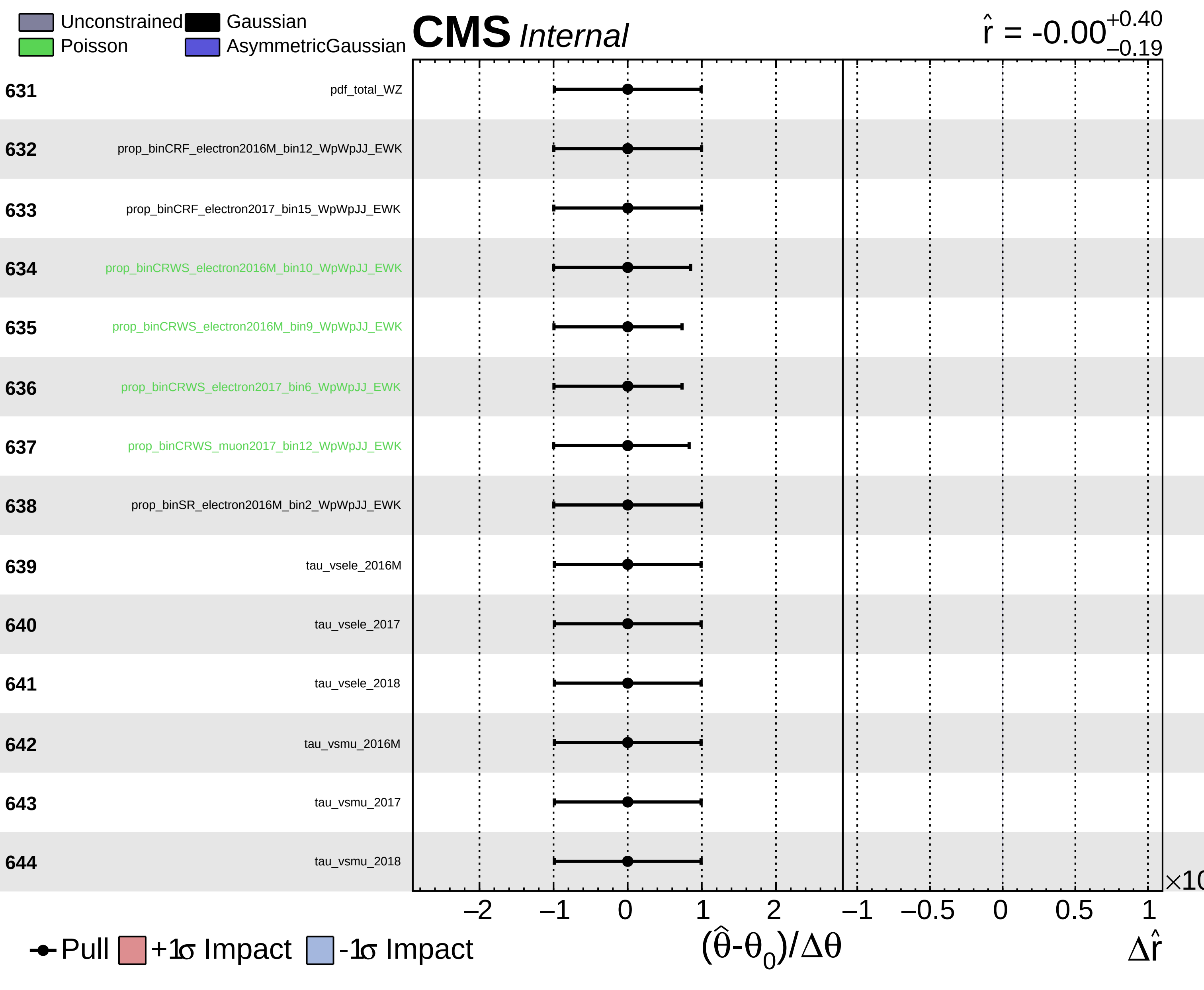
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.40}_{-0.19}$
541 prop_binSR_electron2016M_bin2_WpWpJJ_QCD
542 prop_binCRF_muon2017_bin14_ZZtoLep
543 prop_binCRWS_electron2017_bin11_TTTo2L2Nu
544 prop_binCRWS_electron2016M_bin5_TTTo2L2Nu
545 prop_binSR_electron2016M_bin3_Triboson
546 prop_binCRWS_electron2017_bin6_WZ
547 prop_binCRF_electron2016M_bin12_Triboson
548 prop_binCRWS_electron2017_bin6_WrongSign
549 prop_binSR_electron2016M_bin2_TVX
550 prop_binCRWS_electron2018_bin8_VG
551 prop_binCRF_electron2016M_bin11_ZZtoLep
552 prop_binSR_electron2016M_bin2_Triboson
553 prop_binCRWS_electron2017_bin13_TVX
554 prop_binCRWS_electron2016M_bin12_WrongSign
555 prop_binCRWS_muon2017_bin7_VG
556 prop_binCRWS_electron2016M_bin9_TVX
557 prop_binCRF_electron2016M_bin12_WpWpJJ_QCD
558 prop_binCRWS_electron2016M_bin12_Fake
559 prop_binSR_muon2017_bin6_ZZtoLep
560 prop_binCRWS_electron2016M_bin9_TTTo2L2Nu
561 prop_binCRWS_electron2016M_bin12_Triboson
562 prop_binCRWS_electron2016M_bin5_WrongSign
563 prop_binCRWS_muon2018_bin12_TVX
564 prop_binCRWS_electron2016M_bin10_VG
565 prop_binCRWS_electron2018_bin6_Triboson
566 prop_binCRWS_muon2017_bin12_TVX
567 prop_binCRWS_electron2016M_bin5_WZ
568 prop_binCRWS_electron2018_bin8_Triboson
569 FES
570 prop_binCRWS_electron2018_bin14_Triboson
−2 −1 0 1 2
−5 0 5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = -0.00 +0.40 −0.19
571 prop_binCRWS_muon2017_bin12_Triboson
572 prop_binCRWS_muon2016M_bin10_Triboson
573 prop_binCRWS_muon2018_bin12_Triboson
574 prop_binCRWS_muon2016M_bin10_TVX
575 prop_binCRWS_electron2016M_bin10_TVX
576 prop_binCRWS_electron2017_bin13_Triboson
577 prop_binCRWS_muon2016M_bin10_VG
578 prop_binCRWS_electron2016M_bin12_TVX
579 prop_binCRWS_muon2017_bin12_ZZtoLep
580 prop_binCRWS_electron2016M_bin12_ZZtoLep
581 prop_binCRWS_electron2016M_bin10_ZZtoLep
582 prop_binCRWS_electron2016M_bin10_Triboson
583 prop_binCRWS_electron2017_bin11_ZZtoLep
584 prop_binCRWS_electron2017_bin11_Triboson
585 prop_binCRWS_muon2017_bin7_TVX
586 prop_binCRWS_electron2018_bin14_ZZtoLep
587 prop_binCRWS_electron2017_bin13_ZZtoLep
588 prop_binCRWS_electron2016M_bin5_ZZtoLep
589 prop_binCRWS_electron2018_bin8_ZZtoLep
590 prop_binCRWS_electron2018_bin14_TVX
591 prop_binCRWS_electron2017_bin11_TVX
592 prop_binCRWS_muon2017_bin7_ZZtoLep
593 VBS_2018
594 prop_binCRWS_muon2018_bin5_VG
595 prop_binCRWS_electron2016M_bin9_ZZtoLep
596 prop_binCRWS_electron2018_bin8_TVX
597 prop_binCRWS_electron2018_bin6_ZZtoLep
598 prop_binCRWS_electron2018_bin6_TVX
599 prop_binCRWS_muon2016M_bin10_ZZtoLep
600 prop_binCRWS_muon2018_bin12_ZZtoLep
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.40}_{-0.19}$
601 prop_binCRWS_electron2017_bin6_Triboson
602 prop_binCRWS_electron2017_bin6_TVX
603 prop_binCRWS_electron2017_bin6_ZZtoLep
604 prop_binCRWS_muon2017_bin7_Triboson
605 prop_binCRWS_electron2016M_bin5_TVX
606 prop_binCRWS_muon2018_bin5_TVX
607 prop_binCRWS_muon2018_bin5_Triboson
608 prop_binCRWS_muon2018_bin5_ZZtoLep
609 pdf_total_sig
610 QCDScale_sig
611 prop_binCRWS_electron2016M_bin5_Triboson
612 VBS_2016M
613 prop_binCRWS_electron2017_bin13_WpWpJJ_EWK
614 prop_binCRWS_electron2016M_bin9_Triboson
615 prop_binCRF_electron2017_bin14_WpWpJJ_EWK
616 prop_binSR_electron2016M_bin3_WpWpJJ_EWK
617 prop_binCRF_electron2016M_bin11_WpWpJJ_EWK
618 prop_binCRWS_muon2017_bin7_WpWpJJ_EWK
619 prop_binCRWS_electron2018_bin6_WpWpJJ_EWK
620 prop_binCRWS_muon2016M_bin10_WpWpJJ_EWK
621 prop_binSR_electron2017_bin11_WpWpJJ_EWK
622 prop_binCRWS_electron2018_bin14_WpWpJJ_EWK
623 prop_binCRWS_electron2018_bin8_WpWpJJ_EWK
624 prop_binSR_muon2017_bin6_WpWpJJ_EWK
625 prop_binCRWS_electron2016M_bin12_WpWpJJ_EWK
626 prop_binCRWS_electron2016M_bin5_WpWpJJ_EWK
627 prop_binCRWS_muon2018_bin5_WpWpJJ_EWK
628 prop_binCRF_muon2017_bin14_WpWpJJ_EWK
629 QCDScale_WZ
630 VBS_2017
−2 −1 0 1 2
−0.2 0 0.2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.40}_{-0.19}$
631 pdf_total_WZ
632 prop_binCRF_electron2016M_bin12_WpWpJJ_EWK
633 prop_binCRF_electron2017_bin15_WpWpJJ_EWK
634 prop_binCRWS_electron2016M_bin10_WpWpJJ_EWK
635 prop_binCRWS_electron2016M_bin9_WpWpJJ_EWK
636 prop_binCRWS_electron2017_bin6_WpWpJJ_EWK
637 prop_binCRWS_muon2017_bin12_WpWpJJ_EWK
638 prop_binSR_electron2016M_bin2_WpWpJJ_EWK
639 tau_vsele_2016M
640 tau_vsele_2017
641 tau_vsele_2018
642 tau_vsmu_2016M
643 tau_vsmu_2017
644 tau_vsmu_2018
−2 −1 0 1 2
−1 −0.5 0 0.5 1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$